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BSc THESIS

Score-Based Generative Models for Undersampled MRI Reconstruction: Faithful Reproduction, Uncertainty Quantification, and Domain-Shift Analysis

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ABSTRACT

Accelerated MRI shortens a scan by measuring only part of the data, leaving the rest to be filled in by a reconstruction algorithm. Many different images fit the same measurements, so prior knowledge has to choose among them. Song et al. [1] supply that knowledge with a single score-based generative model [2] whose sampling loop is interleaved, during reconstruction, with a data-consistency step. One trained network therefore serves every acceleration factor, with no retraining. This thesis reimplements the method from scratch in PyTorch and reuses the authors' pretrained prior through a verified weight conversion. It then asks the question the original evaluation leaves open: how far can the method be trusted away from its training data, and what does it cost to restore that trust?

The reimplementation matches the published results to within 0.15 dB. Averaging $S{=}20$ repeated runs improves PSNR by up to 2 dB. The disagreement between runs ranks pixels by their true error, which supports selective review. We then test the prior on clinical brain MRI prepared differently from its training corpus: skulls present, intensity uncalibrated, orientations mixed. The prior's advantage collapses, and at $24\times$ acceleration it becomes *worse than using no model at all* for 94–100% of images across both sources tested, while its own reliability signal weakens. Controlled experiments trace most of this failure to input geometry and intensity scale rather than to the anatomy; at the most extreme acceleration an oracle input fix recovers only part of the loss, so input statistics explain the failure only partially there. Earlier work reports robustness when the anatomy changes under a fixed pipeline; here the same class of solver breaks when the pipeline changes under a fixed anatomy. Constraining the input at test time does not repair the failure. Adapting the model does: three GPU-minutes of fine-tuning restore the prior on every test image ($+5.1$/$+7.1$/$+6.5$ dB at $4/8/24\times$), about forty target-site images are enough, and the adaptation carries over to a source never seen during adaptation, within the same collection. We report all negative results in full; among them, uncertainty-guided acquisition does not beat random selection. The practical implication is that a pretrained prior's gain over zero filling must be re-measured at every new site, and that when it turns negative, minutes of site-local fine-tuning, not input preprocessing, restore it.

SUBJECT AREA: Medical Image Reconstruction

KEYWORDS: score-based generative models, inverse problems, undersampled MRI, uncertainty quantification, domain shift

ΠΕΡΙΛΗΨΗ

Η επιταχυμένη μαγνητική τομογραφία συντομεύει τη σάρωση μετρώντας μόνο ένα μέρος των δεδομένων. Το υπόλοιπο πρέπει να συμπληρωθεί από έναν αλγόριθμο ανακατασκευής. Πολλές διαφορετικές εικόνες ταιριάζουν στις ίδιες μετρήσεις, οπότε η εκ των προτέρων γνώση πρέπει να διαλέξει ανάμεσά τους. Οι Song et al. [1] παρέχουν αυτή τη γνώση με ένα μοντέλο παραγωγής βασισμένο σε σκορ [2]. Κατά την ανακατασκευή, ο δειγματολήπτης του παρεμβάλλεται με ένα βήμα συνέπειας δεδομένων. Έτσι ένα δίκτυο εξυπηρετεί κάθε παράγοντα επιτάχυνσης, χωρίς επανεκπαίδευση. Η παρούσα πτυχιακή υλοποιεί εκ νέου τη μέθοδο σε PyTorch και επαναχρησιμοποιεί το προεκπαιδευμένο prior μέσω επαληθευμένης μετατροπής βαρών. Στη συνέχεια θέτει το ερώτημα που η αρχική αξιολόγηση αφήνει ανοιχτό: πόσο αξιόπιστη είναι η μέθοδος μακριά από τα δεδομένα εκπαίδευσής της, και με ποιο κόστος αποκαθίσταται η αξιοπιστία;

Η υλοποίηση αναπαράγει τα δημοσιευμένα αποτελέσματα εντός $0.15\,$dB. Ο μέσος όρος $S{=}20$ επαναλήψεων βελτιώνει το PSNR έως και $2\,$dB. Η διαφωνία μεταξύ των επαναλήψεων κατατάσσει τα εικονοστοιχεία κατά το πραγματικό τους σφάλμα. Έπειτα δοκιμάζουμε το prior σε κλινική εγκεφαλική MRI με διαφορετική προεπεξεργασία από το σώμα εκπαίδευσης: με κρανίο, χωρίς βαθμονομημένη ένταση, με μικτούς προσανατολισμούς. Το πλεονέκτημα του prior καταρρέει και στο $24\times$ γίνεται *χειρότερο από το να μη χρησιμοποιηθεί κανένα μοντέλο* στο 94–100% των εικόνων και στις δύο πηγές ελέγχου, ενώ εξασθενεί και το ίδιο του το σήμα αξιοπιστίας. Ελεγχόμενα πειράματα αποδίδουν το μεγαλύτερο μέρος της αποτυχίας στη γεωμετρία και στην κλίμακα έντασης της εισόδου και όχι στην ανατομία· στην πιο ακραία επιτάχυνση, μια ιδανική διόρθωση της εισόδου ανακτά μόνο μέρος της απώλειας, οπότε εκεί τα στατιστικά της εισόδου εξηγούν την αποτυχία μόνο εν μέρει. Η βιβλιογραφία αναφέρει ανθεκτικότητα όταν αλλάζει η ανατομία υπό σταθερή αλυσίδα προεπεξεργασίας· εδώ η ίδια κατηγορία επιλυτών σπάει όταν αλλάζει η αλυσίδα υπό σταθερή ανατομία. Ο περιορισμός της εισόδου δεν διορθώνει την αποτυχία. Η προσαρμογή του μοντέλου τη διορθώνει: τρία λεπτά μικρο-προσαρμογής σε GPU αποκαθιστούν το prior σε κάθε εικόνα ελέγχου ($+5.1$/$+7.1$/$+6.5\,$dB στα $4/8/24\times$), περίπου σαράντα εικόνες του νέου κέντρου αρκούν, και η προσαρμογή μεταφέρεται σε πηγή που δεν χρησιμοποιήθηκε ποτέ κατά την προσαρμογή, εντός της ίδιας συλλογής. Αναφέρουμε πλήρως όλα τα αρνητικά αποτελέσματα· μεταξύ αυτών, η καθοδηγούμενη από αβεβαιότητα λήψη δεν υπερτερεί της τυχαίας. Το πρακτικό συμπέρασμα είναι ότι το κέρδος ενός προεκπαιδευμένου prior έναντι της μηδενικής συμπλήρωσης πρέπει να επαναμετράται σε κάθε νέο κέντρο, και ότι, όταν γίνεται αρνητικό, το αποκαθιστούν λίγα λεπτά τοπικής μικρο-προσαρμογής και όχι η προεπεξεργασία της εισόδου.

ΘΕΜΑΤΙΚΗ ΠΕΡΙΟΧΗ: Ανακατασκευή Ιατρικής Εικόνας

ΛΕΞΕΙΣ ΚΛΕΙΔΙΑ: μοντέλα παραγωγής βασισμένα σε σκορ, αντίστροφα προβλήματα, υποδειγματοληπτημένη μαγνητική τομογραφία, ποσοτικοποίηση αβεβαιότητας, μετατόπιση κατανομής

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PREFACE

This thesis was carried out at the Department of Informatics and Telecommunications of the National and Kapodistrian University of Athens, under the supervision of Assistant Professor Vasileios Nakos. All software was written by the author in Python/PyTorch; all experiments were executed on rented NVIDIA GPU instances (a 24 GB RTX 4090 for the reproduction and solver-ablation runs, an A100 for the remaining experiments), with every configuration and result logged to versioned artifact files from which the tables of this document are generated. The complete source code, the experiment logs, and the scripts that regenerate every table and figure accompany the thesis.

1. INTRODUCTION

1.1 The problem: accelerated MRI as an inverse problem

Magnetic Resonance Imaging (MRI) is among the most informative diagnostic imaging modalities and also among the slowest. An MRI scanner does not record an image directly: it records samples of the image's spatial-frequency content, the so-called *k-space*, one line at a time, and the image is obtained from the fully sampled k-space by an inverse Fourier transform. A full acquisition of a single slice requires traversing every k-space line, and a clinical examination consists of many slices in several contrasts, so examinations routinely take tens of minutes. Long scans are expensive, limit patient throughput, are hard on claustrophobic or paediatric patients, and admit motion artefacts.

Why is the acquisition line-by-line at all? MRI encodes position in the phase and frequency of the nuclear-spin signal: each excitation, played against magnetic gradient fields, reads out one trajectory through the spatial-frequency plane. Physics (the repetition time between excitations) sets a floor on how fast the scanner can collect successive lines. Unlike a camera, the scanner cannot simply "expose longer at lower resolution": the time cost is per k-space line, which is why skipping lines is the natural lever for acceleration.

The classical way to shorten a scan is therefore to *measure less*: skip a fraction of the k-space lines and recover the image computationally. If only $1/R$ of the lines are measured, the scan is nominally R times faster (the *acceleration factor*). The price is mathematical: with fewer measurements than unknowns, the reconstruction becomes an ill-posed *linear inverse problem*. Infinitely many images agree exactly with the measured data, and the naive reconstruction (setting every unmeasured frequency to zero and inverse-transforming, so-called *zero filling*) is blurred and aliased. Selecting the right image among all data-consistent candidates requires *prior knowledge* of what plausible images look like. Compressed sensing formalised this insight with sparsity priors [3–5]; the modern successor of that hand-crafted prior is a *learned generative model* of the image distribution.

1.2 The method under study

Song et al. [1] propose a particularly clean instance of the learned-prior approach. They train a single *unconditional* score-based generative model [2, 6] on medical images (a network that learns, at every noise level, in which direction a noisy image should move to look more like the training anatomy). They never train it on the reconstruction task at all. At test time they run the model's usual Predictor–Corrector sampling loop, which turns pure noise into a synthetic image, but interleave it with a *data-consistency* step. At every iteration, the k-space coordinates that were actually measured are overwritten with the measured values (diffused to the current noise level), so the generative process is only ever allowed to invent the *missing* data. Because the measurement enters only through this projection, the same trained model solves *any* linear inverse problem in its image domain, for any acceleration factor and any sampling mask, *without retraining*. This property, one model for every operator, is what makes the approach attractive as infrastructure: a hospital would need one prior per anatomy, not one network per protocol.

The elegance of the construction raises the question this thesis asks: *how far can it be trusted?* The published evaluation is conducted on the same, highly curated dataset the prior was trained on (skull-stripped, intensity-normalised, co-registered brain MRI). A clin-

ically meaningful answer must also cover the images the model will actually meet: scans from other hospitals, other scanners, and other preprocessing pipelines. It must say *how much* quality is lost and *why*, whether the model knows *where* it is wrong, and what the cheapest *repair* is.

The axis along which the distribution shifts turns out to matter as much as its size. Prior work on posterior sampling for compressed sensing argues, theoretically and empirically, that score-based priors are robust to a change of *anatomy* when the acquisition and preprocessing pipeline is held fixed [7]. This thesis measures the complementary axis: the *same* anatomy through a *different pipeline* (skull present, intensity uncalibrated, orientation mixed). The two axes behave very differently: where Jalal et al. report graceful tolerance of a new anatomy under their own protocol (a different prior and sampler), we find that a new pipeline can drive the solver below the zero-filled baseline. Locating a failure on this axis is what makes it cheaply repairable: the fix is a minutes-long adaptation of the prior, not a new model (chapter 7).

1.3 Contributions

This thesis makes the following contributions, each supported by experiments in which every number comes from a logged GPU run.

(i) **A faithful, from-scratch PyTorch reimplementation** of the method of Song et al. [1], whose reference implementation exists only in JAX/Flax. We reimplement the measurement operator, the Variance-Exploding SDE, the Predictor–Corrector sampler and all four data-consistency solver modes, and reuse the authors' pretrained NCSN++ prior through a verified Flax→PyTorch weight conversion (473 parameter tensors matched by name and shape; chapter 3).

(ii) **A quantitative reproduction** of the paper's undersampled-MRI results on BraTS at $4\times$, $8\times$ and $24\times$ acceleration, matching the reported PSNR/SSIM to within 0.15 dB (section 5.1).

(iii) **An uncertainty-guided extension.** Exploiting the sampler's stochasticity, we draw $S{=}20$ posterior samples per measurement, use their per-pixel mean as a Monte-Carlo (MMSE) reconstruction, which improves PSNR by 1.4–2.0 dB over a single sample, and their per-pixel standard deviation as an uncertainty map whose rank correlation with the true error is 0.81–0.87 (section 5.2). We further convert this map into an actionable *selective-prediction* rule (section 7.6).

(iv) **A speed study** relating reconstruction quality to the number of reverse-diffusion steps and to the deterministic probability-flow ODE: 500 steps match 1000 within half a decibel, halving the computational cost (section 5.4).

(v) **A domain-shift analysis**, the core of the thesis (chapters 6 and 7). On real clinical brain MRI from sources independent of the training set, the prior loses 14–24 dB. At $24\times$ acceleration it becomes *actively harmful*, worse than zero filling, for 94–100% of images across both sources tested; at $8\times$ its gain collapses to roughly zero, with a sign that depends on source and protocol. A sequence of controlled experiments (an oracle harmonization study with a pre-specified control arm, a prior-free baseline, and a support-constraint study) attributes most of the failure to input *geometry and intensity scale* rather than to the prior itself; at $24\times$ the oracle input fix recovers only part of the loss, so the attribution there is partial. The same experiments also separate a genuine mechanism from an implementation error of our own that the controls exposed.

(vi) **A practical repair.** Fine-tuning the pretrained prior for 1500 steps ($\approx$ three GPU-minutes) on images from the target collection recovers $+5.1/+7.1/+6.5$ dB at $4/8/24\times$ on 100% of 48 test images ($p{=}7\times10^{-15}$); 40 images suffice (spread 0.22 dB across four set sizes and two seeds); a held-out score-matching loss shows no memorisation signature; and the adaptation *transfers across sources* within the collection ($+8.33$ dB on a source never seen during adaptation; chapter 7).

(vii) **Negative results, reported in full.** An uncertainty-guided *adaptive-acquisition* experiment is reported as a null: guided k-space line selection is statistically indistinguishable from random under a paired test on adequate sample size, and the encouraging preliminary result was noise (section 5.3). A support constraint estimated at test time turns out neutral at best once implemented correctly: a small but significant 0.71 dB cost at $4\times$, with none detectable at $8\times$/$24\times$ (section 7.2).

1.4 Methodological stance

Every reported number comes from a real GPU run with the exact configuration recorded in a machine-readable artifact; no number is estimated, interpolated, or reused across incompatible protocols. Comparisons are *paired* on identical images wherever the design permits, significance is assessed with exact nonparametric tests, and negative or null results are reported with the same prominence as positive ones. Where an experiment has a known confound (and several do), the confound is stated next to the result rather than in a footnote. Scope limits (retrospective simulation, single-coil, noise-free forward model, no clinical validation) are stated explicitly in section 6.6.

1.5 Outline

Chapter 2 develops the mathematical background: linear inverse problems, why one models the *score* rather than the density, denoising score matching, the SDE view of diffusion models, the samplers, and the optimisation machinery, each accompanied by a plain-language reading. Chapter 3 describes the reimplementation, the weight conversion, and our three extensions. Chapter 4 specifies the datasets, the metrics, and the statistical protocol. Chapter 5 reports the reproduction, uncertainty, aggregation, adaptive-acquisition and speed results. Chapter 6 characterises where the method breaks: out-of-distribution modality, independent sources within the same modality, real-tumour fidelity, and synthetic-lesion erasure. Chapter 7 diagnoses the cause and evaluates the repairs, culminating in the fine-tuning result. Chapter 8 places the work in the literature, and chapter 9 concludes.

2. MATHEMATICAL BACKGROUND

This chapter develops, from first principles, every piece of mathematics the method relies on. Each formal statement is followed by a plain-language reading, so that the chapter is accessible to a reader who is comfortable with basic probability and linear algebra but has not met diffusion models before.

2.1 Linear inverse problems and the Bayesian view

Let $\mathbf{x} \in \mathbb{R}^n$ be an unknown image, flattened to a vector ($n = 240 \times 240 = 57{,}600$ throughout this thesis), and let $\mathbf{y} \in \mathbb{R}^m$ be a set of measurements produced by a known linear operator $\mathbf{A} \in \mathbb{R}^{m \times n}$ under additive noise,

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim q_{\boldsymbol{\epsilon}}. \tag{2.1}$$

When $m < n$ the operator maps a high-dimensional space onto a lower-dimensional one, so it cannot be injective: there is an $(n - m)$-dimensional affine subspace of images that produce *exactly* the same measurements. Equation (2.1) is then *ill-posed* in the sense of Hadamard (a solution exists but is not unique), and no algorithm, however clever, can recover $\mathbf{x}$ from $\mathbf{y}$ alone. The missing information must come from somewhere else: from knowledge of which images are *plausible*.

The Bayesian formulation makes this precise. Treating $\mathbf{x}$ as a random variable with prior density $p(\mathbf{x})$, the posterior over images given the measurement is

$$p(\mathbf{x} \mid \mathbf{y}) \;\propto\; q_{\boldsymbol{\epsilon}}(\mathbf{y} - \mathbf{A}\mathbf{x})\, p(\mathbf{x}), \tag{2.2}$$

the product of a *likelihood* term that enforces agreement with the data and a *prior* term that enforces plausibility. Reconstruction becomes the task of summarising this posterior, whether by a sample, by its mean, or by its mode.

In plain words. The measurements pin down part of the image; the rest is genuinely not in the data. The prior supplies the missing part: it encodes what brain MRI "normally looks like", so that among all images agreeing with the measurements, the reconstruction is the one that also looks like anatomy. Everything in this thesis is a story about what happens when that prior is right and when it is wrong.

The MRI measurement operator. For single-coil Cartesian MRI, the scanner measures samples of the image's 2-D Fourier transform. Writing $\mathbf{T}$ for the (orthonormalised) 2-D discrete Fourier transform and $\boldsymbol{\Lambda} \in \{0, 1\}^{n \times n}$ for a diagonal binary mask that selects which k-space coordinates are measured, the operator factorises as

$$\mathbf{A} = \mathcal{P}(\boldsymbol{\Lambda})\, \mathbf{T}, \tag{2.3}$$

where $\mathcal{P}(\boldsymbol{\Lambda})$ simply drops the coordinates zeroed by $\boldsymbol{\Lambda}$. In Cartesian undersampling the mask keeps whole k-space *columns*: a fully sampled block at the centre (low frequencies, which carry contrast) and equispaced columns elsewhere. The acceleration factor R means a fraction $1/R$ of columns is kept. Changing R changes only $\boldsymbol{\Lambda}$; the transform

$\mathbf{T}$, and with it everything the generative model learned, is untouched. This factorisation, Proposition 1 of Song et al. [1], is what lets one model serve every acceleration. One typing remark: the DFT is complex-valued, so for MRI the measurements live in $\mathbb{C}^m$ and $\mathbf{A} \in \mathbb{C}^{m \times n}$ (the image itself remains real, since we work with magnitude images); identifying $\mathbb{C}^m$ with $\mathbb{R}^{2m}$ and replacing transposes by conjugate transposes, everything in section 2.1 carries over verbatim, and we keep the real-valued notation, as does [1].

Classical solutions. The pseudo-inverse reconstruction $\hat{\mathbf{x}} = \mathbf{A}^{+}\mathbf{y}$ is the *minimum-norm* least-squares solution: with $m < n$ infinitely many images fit the data exactly, and $\mathbf{A}^{+}\mathbf{y}$ selects the smallest of them. For the operator in eq. (2.3), whose rows are orthonormal, this coincides with *zero filling*: set every unmeasured coordinate to zero and apply $\mathbf{T}^{-1}$. It is the natural baseline in the absence of a model (though even the choice to zero the unmeasured coordinates is itself a minimal prior), and it is the reference baseline throughout this thesis ("what you get with no model at all"). Compressed sensing [3–5] improves on it by adding a sparsity prior (e.g. minimising the ℓ_1 norm of the wavelet coefficients subject to data consistency), with recovery guarantees when the image is compressible and the sampling sufficiently incoherent. The method studied here replaces the hand-crafted sparsity prior with a learned generative model; the data-consistency machinery remains recognisably the same.

2.2 Why the score, not the density

A generative prior must represent a probability density over images. Modelling the density directly runs into a wall. Any energy-based parameterisation

$$p_\theta(\mathbf{x}) = \frac{e^{-E_\theta(\mathbf{x})}}{Z_\theta}, \qquad Z_\theta = \int e^{-E_\theta(\mathbf{x})}\, \mathrm{d}\mathbf{x}, \tag{2.4}$$

requires the normalising constant Z_θ, an integral over *all* 240×240 images (i.e. over $\mathbb{R}^{57{,}600}$), which is intractable and, worse, changes with every update of θ.

The escape, due to score matching [8], is to model not the density but its *score*: the gradient of the log-density with respect to the image,

$$\nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}) = -\nabla_{\mathbf{x}} E_\theta(\mathbf{x}) \qquad (\text{because } \nabla_{\mathbf{x}} \log Z_\theta = 0). \tag{2.5}$$

Differentiation with respect to $\mathbf{x}$ kills the term that made eq. (2.4) intractable, because Z_θ does not depend on $\mathbf{x}$. The score is a *vector field*: at every image it is an image-shaped arrow pointing in the direction of steepest increase of log-probability.

In plain words. We give up on asking "how probable is this image?" (the number that requires the impossible integral) and instead ask "in which direction should each pixel move for the image to become more probable?", a question whose answer never involves the integral. That directional information turns out to be all a sampler needs.

2.3 Denoising score matching

The score of the *data* distribution is unknown, so it cannot be regressed on directly. Denoising score matching (DSM) [9] provides a supervised surrogate. Perturb a data point

$\mathbf{x}_0 \sim p_{\text{data}}$ with Gaussian noise of known scale σ,

$$\tilde{\mathbf{x}} = \mathbf{x}_0 + \sigma \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{2.6}$$

and denote by p_σ the resulting noisy marginal. For the Gaussian perturbation kernel the conditional score has closed form,

$$\nabla_{\tilde{\mathbf{x}}} \log p(\tilde{\mathbf{x}} \mid \mathbf{x}_0) = -\frac{\tilde{\mathbf{x}} - \mathbf{x}_0}{\sigma^2} = -\frac{\mathbf{z}}{\sigma}, \tag{2.7}$$

so a network s_θ can be trained to predict it. The DSM objective, in the noise-weighted form used by the model studied here [2, 6], is

$$L(\theta) \;=\; \mathbb{E}_{t \sim \mathcal{U}(t_{\text{min}},1)}\, \mathbb{E}_{\mathbf{x}_0}\, \mathbb{E}_{\mathbf{z}} \Big\| \, \sigma(t)\, s_\theta\big(\mathbf{x}_0 + \sigma(t)\mathbf{z},\; t\big) + \mathbf{z} \, \Big\|_2^2, \tag{2.8}$$

where the truncation $t_{\text{min}} = 10^{-5}$ follows the reference implementation (under the convention of eq. (2.13) the limit $t \to 0$ is in fact benign, since $\sigma(0) = \sigma_{\text{min}} > 0$ keeps the target bounded; the truncation matters for schedules whose perturbation variance vanishes at $t = 0$), $\sigma(t)$ is the noise schedule of section 2.4, and the factor $\sigma(t)$ inside the norm keeps the regression target of order one at every noise level, so that no level dominates the loss.

Two facts make eq. (2.8) remarkable. First, it is *self-supervised*: because the noise $\mathbf{z}$ is added by us, the regression target is known exactly, so training needs no labels and no clean/corrupted pairs from a scanner, only a pile of images. Second, its minimiser is exactly the score of the noisy marginal. Writing the optimal predictor of the conditional target as a conditional expectation,

$$s_{\theta^\star}(\tilde{\mathbf{x}}, t) = \mathbb{E}\left[-\frac{\mathbf{z}}{\sigma(t)} \;\middle|\; \tilde{\mathbf{x}} \right] = \nabla_{\tilde{\mathbf{x}}} \log p_{\sigma(t)}(\tilde{\mathbf{x}}), \tag{2.9}$$

where the second equality is Tweedie's identity / the result of Vincent [9]. Note the subtlety: the minimiser is *not* $-\mathbf{z}/\sigma$ itself (many different noise draws produce the same noisy image $\tilde{\mathbf{x}}$), but the *average* of that quantity over all of them; that average is precisely the score of the noise-perturbed density p_σ, not of the clean density.

Why the minimiser is the score: a two-line derivation. Both facts of eq. (2.9) follow from elementary manipulations worth making explicit. For any square loss $\mathbb{E}\|f(\tilde{\mathbf{x}}) - \mathbf{t}\|^2$ over pairs $(\tilde{\mathbf{x}}, \mathbf{t})$, the minimising function is the conditional mean $f^\star(\tilde{\mathbf{x}}) = \mathbb{E}[\mathbf{t} \mid \tilde{\mathbf{x}}]$, which gives the first equality. For the second, write the noisy marginal as the convolution $p_\sigma(\tilde{\mathbf{x}}) = \int p_{\text{data}}(\mathbf{x}_0)\, \mathcal{N}(\tilde{\mathbf{x}}; \mathbf{x}_0, \sigma^2 \mathbf{I})\, \mathrm{d}\mathbf{x}_0$ and differentiate under the integral:

$$\nabla_{\tilde{\mathbf{x}}} \log p_\sigma(\tilde{\mathbf{x}}) = \frac{\int p_{\text{data}}(\mathbf{x}_0)\, \mathcal{N}(\tilde{\mathbf{x}}; \mathbf{x}_0, \sigma^2 \mathbf{I}) \cdot \frac{\mathbf{x}_0 - \tilde{\mathbf{x}}}{\sigma^2}\, \mathrm{d}\mathbf{x}_0}{p_\sigma(\tilde{\mathbf{x}})} = \mathbb{E}\left[\frac{\mathbf{x}_0 - \tilde{\mathbf{x}}}{\sigma^2} \;\middle|\; \tilde{\mathbf{x}} \right] = \mathbb{E}\Big[-\frac{\mathbf{z}}{\sigma} \;\Big|\; \tilde{\mathbf{x}} \Big], \tag{2.10}$$

using $\nabla_{\tilde{\mathbf{x}}} \mathcal{N}(\tilde{\mathbf{x}}; \mathbf{x}_0, \sigma^2 \mathbf{I}) = \mathcal{N}(\tilde{\mathbf{x}}; \mathbf{x}_0, \sigma^2 \mathbf{I}) \cdot (\mathbf{x}_0 - \tilde{\mathbf{x}})/\sigma^2$ and $\tilde{\mathbf{x}} = \mathbf{x}_0 + \sigma \mathbf{z}$. The denoising regression and the score are therefore the same object; no approximation is involved beyond the network's capacity.

The network. The score model used throughout is NCSN++ [2], a time-conditional U-Net [10]: an encoder–decoder convolutional network with residual blocks, group normalisation, and skip connections between resolutions, in which the noise level t enters through

a Fourier-feature embedding added to every residual block. The instance reused here (trained by [1] on BraTS) takes one 240×240 channel and has $2{,}676{,}069$ parameters, small by modern standards. This matters twice in this thesis: it makes the ≈ 2000 network evaluations of one reconstruction affordable, and it makes the three-minute fine-tuning of chapter 7 possible.

In plain words. Training is a denoising exercise: show the network a noisy brain scan, ask it to point back toward the clean one. Because we corrupted the image ourselves, we can grade the answer exactly. A network that wins this game has implicitly learned, at every noise level, which way "more like a brain" lies, and that direction is the score. The prior used in this thesis is such a network (NCSN++ [2]), trained by Song et al. [1] on brain MRI; this thesis never trains it from scratch, but does *fine-tune* it in chapter 7.

2.4 Diffusion: the SDE view

Score-based generative modelling turns the static denoiser of section 2.3 into a generative process by embedding it in a continuous-time diffusion [2]. A forward stochastic differential equation (SDE) over $t \in [0, 1]$ gradually destroys structure,

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{w}, \tag{2.11}$$

with drift $\mathbf{f}$, diffusion coefficient g, and $\mathbf{w}$ a standard Wiener process. This thesis uses the *Variance-Exploding* (VE) instance, the continuous limit of the NCSN noise ladder [6]:

$$\mathbf{f} \equiv \mathbf{0}, \qquad g(t) = \sqrt{\frac{\mathrm{d}[\sigma^2(t)]}{\mathrm{d}t}}, \qquad \sigma(t) = \sigma_{\mathsf{min}} \left(\frac{\sigma_{\mathsf{max}}}{\sigma_{\mathsf{min}}}\right)^{t}, \tag{2.12}$$

with $\sigma_{\mathsf{min}} = 0.01$ and $\sigma_{\mathsf{max}} = 128$ for the model at hand. The name "variance exploding" is apt: the image is never scaled down, noise of geometrically growing standard deviation is simply piled on top, until at $t = 1$ the noise (std 128 on images valued in $[0, 1]$) completely swamps the content. The associated perturbation kernel is Gaussian. Integrating the SDE exactly gives variance $[\sigma^2(t) - \sigma^2_{\mathsf{min}}]\,\mathbf{I}$, and since $\sigma^2_{\mathsf{min}} = 10^{-4}$ is negligible the standard convention, used by the training objective and by the implementation alike, absorbs it:

$$p_{0t}(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_t;\ \mathbf{x}_0,\ \sigma^2(t)\,\mathbf{I}\big), \qquad \text{i.e.} \quad \mathbf{x}_t = \mathbf{x}_0 + \sigma(t)\,\mathbf{z}, \tag{2.13}$$

which is the perturbation used in the DSM objective eq. (2.8); the training loss and the generative dynamics fit together by construction. (The exact solution of eq. (2.11) has variance $\sigma^2(t) - \sigma^2_{\mathsf{min}}$; following the reference implementation we use $\sigma^2(t)$. The noise-amplitude discrepancy $\sigma(t) - \sqrt{\sigma^2(t) - \sigma^2_{\mathsf{min}}}$ is largest at $t = 0$, where it equals $\sigma_{\mathsf{min}} = 0.01$, negligible against the $[0, 1]$ image range.)

Reverse-time SDE. The classical result of Anderson [11] states that the time reversal of eq. (2.11) is again a diffusion,

$$\mathrm{d}\mathbf{x} = \big[\mathbf{f}(\mathbf{x}, t) - g(t)^2\,\nabla_{\mathbf{x}} \log p_t(\mathbf{x})\big]\,\mathrm{d}t + g(t)\,\mathrm{d}\bar{\mathbf{w}}, \tag{2.14}$$

run backwards from $t = 1$ to $t = 0$, where $\bar{\mathbf{w}}$ is a reverse-time Wiener process. The reverse dynamics depend on the data distribution through *one* quantity only: the score $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ of the noisy marginals, which is what DSM estimates. Substituting $s_\theta \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ into

eq. (2.14) and integrating numerically from Gaussian noise produces samples from (approximately) the data distribution.

Probability-flow ODE. The same marginals are shared by a deterministic ordinary differential equation [2],

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \mathbf{f}(\mathbf{x}, t) - \tfrac{1}{2}\, g(t)^2\, \nabla_{\mathbf{x}} \log p_t(\mathbf{x}), \tag{2.15}$$

which removes the injected noise entirely; we use it in the speed study of section 5.4 as an alternative predictor.

Langevin corrector. At a *fixed* noise level t, samples can also be refined by score-based Langevin MCMC,

$$\mathbf{x} \leftarrow \mathbf{x} + \varepsilon\, s_\theta(\mathbf{x}, t) + \sqrt{2\varepsilon}\, \mathbf{z}, \qquad \varepsilon = 2\left(\frac{\eta\, \|\mathbf{z}\|_2}{\|s_\theta(\mathbf{x}, t)\|_2}\right)^2, \tag{2.16}$$

where the step size ε is set adaptively from a target signal-to-noise ratio η (here the paper's tuned $\eta = 0.577$; the norms are batch means, so the step is shared across a batch; in the general PC algorithm a VP-specific factor α_i multiplies this expression, which equals 1 for the VE-SDE used here). Langevin steps do not advance time; they correct the distributional error a discrete time step leaves behind.

From continuous to discrete: the sampler as an algorithm. In practice eq. (2.14) is discretised on a geometric grid of noise levels $\sigma_{\mathsf{min}} = \sigma_0 < \sigma_1 < \cdots < \sigma_{N-1} = \sigma_{\mathsf{max}}$ (the discrete counterpart of the schedule in eq. (2.12)), with the boundary convention $\sigma_{-1} := 0$ so that the final step is defined. Starting from $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathsf{max}}^2 \mathbf{I})$, one iteration at level i performs:

1. **Predictor** (the reverse-diffusion predictor of Song et al. [2] for the VE-SDE): with $G_i^2 = \sigma_i^2 - \sigma_{i-1}^2$,

$$\mathbf{x} \;\leftarrow\; \mathbf{x} + G_i^2\, s_\theta(\mathbf{x}, \sigma_i) + G_i\, \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{2.17}$$

   which is eq. (2.14) discretised on the grid of the forward chain ($\mathbf{f} = \mathbf{0}$): the injected variance $G_i^2 = \int_{t_{i-1}}^{t_i} g^2(s)\, \mathrm{d}s$ integrates $g^2\, \mathrm{d}t$ over the step exactly, so this is the Euler–Maruyama scheme after the time change $\tau = \sigma^2(t)$; a plain Euler–Maruyama step in t would use $g(t_i)^2/N$ instead and agrees to first order. (SMLD ancestral sampling is a distinct predictor whose noise coefficient carries an extra factor σ_{i-1}/σ_i, negligible for large N.) In effect: a deterministic move along the score plus an injection of fresh noise at the reduced level.

2. **Corrector**: one Langevin step (eq. (2.16)) at the new, fixed level, repairing the distributional error the finite step introduced.

With N levels and one corrector step per level, a reconstruction costs about $2N$ score-network evaluations (≈ 2000 at the default $N = 1000$), which is the quantity the speed study of section 5.4 minimises, and which the 20-sample uncertainty method of section 3.3 multiplies twenty-fold. The reconstruction algorithm of chapter 3 inserts one further operation, the data-consistency projection, before each predictor–corrector pair.

In plain words. The forward process is a recipe for destroying an image with noise, chosen so that (a) we can jump to any intermediate noise level in one step (eq. (2.13)), and (b) the process can be run *backwards* if one knows the score. Sampling is the backward run: start from pure noise, repeatedly take a small denoising step (predictor) followed by a small cleanup step (corrector), and after N rounds an image of the learned anatomy emerges. The continuous-time (SDE) formulation is what makes the noise level a continuous dial, so a single network serves every level and the number of steps becomes a free choice at sampling time, a fact exploited in section 5.4.

2.5 Optimisation machinery

The prior was trained, and in chapter 7 is fine-tuned, by stochastic gradient methods, which we summarise for completeness, since the fine-tuning result depends on these details being kept identical to the original recipe.

Plain gradient descent updates parameters θ against the gradient of the loss, $\theta \leftarrow \theta - \eta_{\text{lr}} \nabla_\theta L(\theta)$, with learning rate η_{lr}. For losses that are expectations over data, the gradient is estimated on mini-batches (stochastic gradient descent), which is computationally necessary but noisy. Adam [12] stabilises it with two per-parameter accumulators: momentum $m \leftarrow \beta_1 m + (1-\beta_1) g$ smooths successive gradients, and a second moment $v \leftarrow \beta_2 v + (1-\beta_2) g^2$ rescales each coordinate by its own gradient magnitude. These are combined with bias corrections $\hat{m} = m/(1-\beta_1^t)$, $\hat{v} = v/(1-\beta_2^t)$ that undo the zero-initialisation bias of the accumulators (we write Adam's damping constant as δ to keep ε for the Langevin step size of eq. (2.16)):

$$\theta \leftarrow \theta - \eta_{\text{lr}} \frac{\hat{m}}{\sqrt{\hat{v}} + \delta}. \tag{2.18}$$

The training recipe of the prior (kept identical in our fine-tuning; Annex I) is Adam with $\eta_{\text{lr}} = 2 \times 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, a *linear warm-up* of the learning rate over the first 5000 steps (early on, $\hat{v}$ is estimated from very few gradients, so the per-coordinate scaling is unreliable and a small learning rate limits the damage), *gradient clipping* to global ℓ_2 norm 1 (rare batches produce outlier gradients whose influence must be bounded), and an *exponential moving average* (EMA) of the weights with decay 0.999. The EMA is evaluated in place of the last iterate (the average of the recent trajectory generalises better than its endpoint). Because the fine-tuning run of chapter 7 lasts only 1500 steps and warm-up restarts, its learning rate ramps to $\approx 6 \times 10^{-5}$ rather than reaching the nominal 2×10^{-4}, and the EMA decay cap $(1+n)/(10+n)$ makes the effective decay ≈ 0.994. These details matter when reproducing the numbers.

In plain words. Training walks downhill on the loss surface in small steps. Adam makes the walk robust in two ways: it averages the direction over recent steps so the path stops zig-zagging, and it gives every parameter its own step size so that rarely-updated parameters still move. The three extras are guard rails: start gently while the step-size statistics are still unreliable (warm-up), never let a single freak batch yank the parameters (clipping), and judge the model by the average of where it has recently been rather than by its very last position (EMA). None of this is exotic; what matters for this thesis is that the fine-tuning of chapter 7 inherits the recipe unchanged, so its gains cannot be attributed to optimisation tricks.

2.6 Estimation-theoretic facts used later

Two standard results anchor the uncertainty extension of section 5.2. First, the estimator minimising the expected squared error $\mathbb{E}\|\mathbf{x} - \hat{\mathbf{x}}\|^2$ given the measurement $\mathbf{y}$ is the *posterior mean* $\hat{\mathbf{x}} = \mathbb{E}[\mathbf{x} \mid \mathbf{y}]$ (the MMSE estimator); a Monte-Carlo average of posterior samples is a consistent estimate of it. Second, the posterior mean is smoother than typical posterior samples for an elementary reason: averaging cancels the high-frequency detail on which the samples disagree, so structure the data do not pin down is attenuated rather than invented. Blau and Michaeli [13] formalise the general phenomenon as the *perception–distortion trade-off*: any estimator with sufficiently low distortion must have an output *distribution* measurably different from that of real images. A true posterior sample, by contrast, is distributed exactly like real images and sits at the other end of the trade-off. The reading we adopt when interpreting table 5.2 is that the posterior mean scores higher PSNR precisely *because* it is smoother where the data are uninformative, not despite it. A single posterior sample is the sharper, more “photorealistic” object.

3. METHOD AND IMPLEMENTATION

This chapter describes what was built: the faithful PyTorch reimplementation of the solver of Song et al. [1] (sections 3.1 and 3.2), and the three extensions this thesis adds on top of it, namely uncertainty-guided reconstruction (section 3.3), uncertainty-guided adaptive acquisition (section 3.4), and the sampling-speed study (section 3.5).

3.1 PyTorch reimplementation and verified weight conversion

The authors of [1] released a JAX/Flax implementation. We reimplement the method from scratch in PyTorch. The unconditional backbone (the NCSN++ architecture, the VE-SDE of eq. (2.12) and the Predictor–Corrector sampler) follows the reference PyTorch port of [2]. The measurement operator $\mathbf{A} = \mathcal{P}(\Lambda)\mathbf{T}$ of eq. (2.3) (2-D FFT/IFFT with orthonormal scaling, Cartesian mask construction, and the complex-number book-keeping), the noised-measurement process, the data-consistency step, and all four solver modes are our own PyTorch code, ported statement-by-statement from the reference. CPU unit tests validate three properties: the FFT round-trip ($\mathbf{T}^{-1}\mathbf{T} = \mathbf{I}$ to machine precision); the mask construction (fully sampled low-frequency centre plus equispaced columns, matching the reference's actual sampled-column counts of $60/31/10$ at $4/8/24\times$ for nominal $n_{\text{keep}} = 60/30/10$); and exactness of the data-consistency step at $\lambda = 1$ ($\mathbf{A}\hat{\mathbf{x}}' = \hat{\mathbf{y}}$ after projection).

We do *not* retrain the prior. Instead we convert the authors' pretrained Flax NCSN++ checkpoint (the exponential-moving-average weights, `checkpoint_26`) into a PyTorch `state_dict`. The conversion is name-based: each PyTorch parameter `all_modules.`i`....` is matched to its Flax counterpart `<Class>_`k`/...` (per-class creation-order indices), applying the axis transforms Conv HWIO $\to$ OIHW, Dense $(\text{in}, \text{out}) \to (\text{out}, \text{in})$, and GroupNorm $(1, 1, 1, C) \to (C)$. All 473 parameter tensors match with identical shapes, and the converted model loads with zero missing or unexpected keys; the network has $2{,}676{,}069$ parameters in total. This procedure obtains the paper's reconstruction quality at zero training cost. Because the reproduction of section 5.1 agrees with the published numbers to within 0.15 dB, it also validates the conversion and the reimplementation end-to-end: an error in either would surface as a quality gap.

In plain words. We rebuilt the machine in a different framework and transplanted the original brain into it. The transplant is checked bolt-by-bolt (every one of the 473 weight tensors is matched by name and shape), and then the whole machine is checked by racing it against the original: it produces the same results to within measurement noise.

3.2 Conditioning: the data-consistency solver

Reconstruction runs the unconditional PC sampler of section 2.4 and interleaves it with a data-consistency step, following [1]. Because the sampler's iterate at time t is a *noisy* image (eq. (2.13)), the measurement must first be brought to the same noise level:

$$\hat{\mathbf{y}}_t = \mathbf{y} + \sigma(t)\,\mathbf{A}\mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{3.1}$$

Splicing the *clean* measurement into a noisy iterate would create an image unlike anything the score network has ever seen; eq. (3.1) keeps the spliced-in data statistically indistin-

guishable from the rest of the iterate. (Exactly this consideration, ignored in one of our own later experiments, produced a bug that the controls of section 7.2 caught; the lesson generalises.)

The data-consistency step itself (Theorem 1 of [1]), applied before each predictor/corrector update with slack $0 \leq \lambda \leq 1$, is

$$\hat{\mathbf{x}}_t' = \mathbf{T}^{-1}\big[\, \lambda\, \boldsymbol{\Lambda}\, \mathcal{P}^{-1}(\boldsymbol{\Lambda})\, \hat{\mathbf{y}}_t + (1-\lambda)\, \boldsymbol{\Lambda}\, \mathbf{T}\hat{\mathbf{x}}_t + (\mathbf{I} - \boldsymbol{\Lambda})\, \mathbf{T}\hat{\mathbf{x}}_t \,\big]. \tag{3.2}$$

Read in the transform domain: on the *measured* k-space coordinates, replace the current estimate by a λ-weighted blend of the (noised) measurement and the estimate; on the *unmeasured* coordinates, change nothing. Setting $\lambda = 0$ recovers unconditional sampling; $\lambda = 1$ enforces $\mathbf{A}\hat{\mathbf{x}}_t' = \hat{\mathbf{y}}_t$ exactly. Following the paper's tuned optimum we use $\lambda = 0.982$ during sampling and $\lambda = 1$ at the final denoising step, so the output agrees exactly with the measurements. The corrector SNR is the paper's $\eta = 0.577$.

[1] define four solver modes; we reimplement all four. `projection` (the default everywhere in this thesis) interleaves eq. (3.2) with the full PC sampler; `langevin_projection` ("ALD+Ours") uses annealed Langevin dynamics as the sampler; `langevin` and `baseline` are the paper's two baselines without the data-consistency projection.

Putting sections 3.1 and 3.2 together, one reconstruction runs as follows.

1. Simulate (or receive) the measurement $\mathbf{y} = \mathbf{A}\mathbf{x}^{\text{gt}}$ with mask $\boldsymbol{\Lambda}$.
2. Initialise $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{max}}^2 \mathbf{I})$.
3. For $i = N-1$ down to 0: (a) draw $\hat{\mathbf{y}}_{t_i}$ by eq. (3.1); (b) project $\mathbf{x}$ by eq. (3.2) with $\lambda = 0.982$; (c) predictor step eq. (2.17); (d) corrector step eq. (2.16).
4. Final denoise with $\lambda = 1$, so the output satisfies $\mathbf{A}\hat{\mathbf{x}} = \mathbf{y}$ exactly; clip to $[0, 1]$.

Steps (a)–(d) cost two score evaluations, so the full run costs $\approx 2N$; everything except the projection (b) and the noising (a) is the *unchanged* unconditional sampler, which is the practical meaning of "one prior, every operator".

Correctness testing. Faithfulness was enforced at three levels: deterministic CPU unit tests on the measurement operator, shape-exact conversion checks on the checkpoint, and the end-to-end reproduction of section 5.1. Each is detailed above where the step it validates is introduced. For the default `projection` path (operator, weight conversion, full sampler) no subtle porting error could survive all three. The three alternative solver modes are validated only by the coarser sanity comparison of section 5.1, which indeed flagged one unresolved discrepancy, the plain Langevin baseline.

In plain words. The generative model would happily dream up *any* brain. The data-consistency step is a leash: at every one of the 1000 steps, whatever the scanner actually measured is written back over the dream. The model is left free to invent only what was never measured, which is exactly the part that must be invented.

3.3 Extension 1: uncertainty-guided reconstruction

The solver is stochastic three times over: the initial iterate is random noise, the corrector injects Langevin noise, and the measurement is re-noised by eq. (3.1) at every step. Re-

peated runs on the *same* measurement $\mathbf{y}$ therefore draw different samples $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(S)}$ from (an approximation of) the posterior $p(\mathbf{x} \mid \mathbf{y})$ (we write S for the number of posterior samples, reserving N for the number of reverse-diffusion steps). Rather than treat this stochasticity as a nuisance, we aggregate it per pixel:

$$\bar{\mathbf{x}} = \frac{1}{S}\sum_{j=1}^{S} \mathbf{x}^{(j)}, \qquad \mathbf{u} = \Big(\frac{1}{S}\sum_{j=1}^{S} (\mathbf{x}^{(j)} - \bar{\mathbf{x}})^2\Big)^{1/2}. \tag{3.3}$$

By section 2.6, $\bar{\mathbf{x}}$ is a Monte-Carlo estimate of the posterior mean (the MMSE reconstruction), and $\mathbf{u}$ is the per-pixel posterior spread: where the S runs agree, the measurement pinned the answer down; where they disagree, the model was guessing. We use $S = 20$ for the in-distribution experiments of chapter 5; the larger sweeps and repair studies of chapters 6 and 7 reduce S (to 8, 6 or 4) to keep compute bounded, with the value stated in every table caption. The choice reflects a trade-off: Monte-Carlo error shrinks as $1/\sqrt{S}$, so the first handful of samples buys most of the averaging benefit while the uncertainty map needs enough samples for a stable per-pixel standard deviation. $S = 20$ balances both against a $20\times$ compute bill. As robustness checks we also evaluate two alternatives to the mean: the per-pixel *median*, and a *trimmed mean* discarding the $\lfloor S/4 \rfloor$ lowest and highest values per pixel.

Data-consistency filter. Before aggregating, samples inconsistent with the measurement are discarded by a robust rule on the relative residual $r^{(j)} = \|\mathbf{\Lambda T x}^{(j)} - \mathcal{P}^{-1}(\mathbf{\Lambda})\mathbf{y}\|/\|\mathbf{y}\|$ (with $\mathcal{P}^{-1}(\mathbf{\Lambda})\mathbf{y}$ the zero-filled embedding of $\mathbf{y}$ into $\mathbb{R}^n$): sample j is kept iff $r^{(j)} \le 1.02 \cdot \mathrm{med}_j\, r^{(j)} + 3 \cdot \mathrm{MAD}_j\, r^{(j)}$ (median absolute deviation, plus a 2%-of-median slack that prevents over-pruning when the residuals cluster so tightly that the MAD is near zero); as a defensive floor, the two lowest-residual samples are always retained. For the `projection` solver this retains essentially all samples (mean kept ≈ 19.9 of $S = 20$); it exists to guard the aggregate against the occasional diverged run.

Calibration. An uncertainty map is only useful if it is *informative about the true error*. We measure, per image over the evaluation region, the Pearson and Spearman (rank) correlations between the predicted uncertainty u_i and the realised absolute error $e_i = |\bar{x}_i - x_i^{\mathrm{gt}}|$, and report their averages. A Spearman correlation near 1 means the map correctly *ranks* pixels from most to least trustworthy. We emphasise the interpretation: this is ranking quality, not a calibrated error probability: the map says “check here first”, not “this pixel is wrong with probability p”. Section 7.6 builds a decision rule on exactly this ranking property.

In plain words. We ask the model for twenty independent answers to the same question. Their average is a better answer than any single one (noise cancels), and their disagreement is a confession: wherever the twenty answers differ, the data did not determine the answer and the model filled the gap. That confession map is the safety feature this thesis measures and then puts to work.

3.4 Extension 2: uncertainty-guided adaptive acquisition

Because the mask Λ can change at test time without retraining, the uncertainty of eq. (3.3) suggests a closed-loop idea: measure a first pass, see where the model is uncertain, and spend the next measurements *there*. From K posterior samples (a smaller draw than S;

here $K = 8$) we compute a per-column importance score: the cross-sample variance of the k-space content, averaged over rows,

$$\mathrm{imp}(c) = \frac{1}{HK} \sum_{r,j} \left| (\mathbf{T}\mathbf{x}^{(j)})_{r,c} - \overline{(\mathbf{T}\mathbf{x})}_{r,c} \right|^2, \tag{3.4}$$

and add the top-B currently *missing* columns to the mask (strategy `guided`). Two controls receive the same budget B: `random` adds B random missing columns, and `lowfreq` adds the B missing columns nearest the k-space centre. All three arms then reconstruct again with their enriched masks, and `guided` is compared against `random` with a *paired* per-slice test, the design that section 5.3 shows to be decisive.

3.5 Extension 3: sampling speed

The default sampler uses $N = 1000$ reverse-diffusion steps, i.e. ≈ 2000 score evaluations per sample; the uncertainty method of section 3.3 multiplies that by 20. We therefore measure reconstruction quality as a function of $N \in \{1000, 500, 200, 100, 50\}$, and separately compare the stochastic reverse-diffusion predictor against the deterministic probability-flow ODE predictor (eq. (2.15)) at equal N. Cost is reported in score evaluations rather than seconds, because wall-clock at small N is dominated by fixed per-run overhead (model construction, CUDA initialisation, data loading) that does not scale with N.

4. DATA, METRICS AND STATISTICAL PROTOCOL

4.1 Datasets

Training data of the prior (BraTS 2021). The pretrained NCSN++ prior we reuse was trained by Song et al. [1] on the *RSNA–ASNR–MICCAI BraTS 2021* brain-tumour dataset [14–16]. BraTS 2021 comprises $2{,}000$ cases ($8{,}000$ multi-parametric MRI scans) with four co-registered contrasts per case: native T1, post-contrast T1 (T1Gd), T2, and T2-FLAIR. The volumes are resampled to $1\,\mathrm{mm}^3$ isotropic resolution and *skull-stripped*, giving volumes of $240 \times 240 \times 155$ voxels. The authors slice the 3-D volumes into $297{,}270$ two-dimensional images of size 240×240 for training. Three properties of this corpus matter for everything that follows: the images contain *no skull* (the brain floats on a black background; 73.2% of BraTS pixels are exactly zero), the intensity scale is normalised per volume, and the geometry is standardised by co-registration. Any image without these properties is, strictly, outside the training distribution, which is the crux of chapters 6 and 7. Since BraTS is resampled to $1\,\mathrm{mm}^3$, one pixel corresponds to one millimetre, a correspondence we use when giving lesion sizes in section 6.5.

Evaluation data (in-distribution). We evaluate on the authors' released test split `BraTS.npz`: 1000 magnitude slices of size 240×240, stored as `uint8` in $[0, 255]$ and scaled to $[0, 1]$. Single-coil k-space is simulated by a 2-D FFT and undersampled with an equispaced Cartesian mask (fully sampled low-frequency centre) at acceleration factors $R \in \{4, 8, 24\}$; the nominal kept-column parameter is $n_{\text{keep}} = 60/30/10$ respectively (the resulting masks contain $60/31/10$ sampled columns; at $8\times$ the union of the fully sampled centre and the equispaced columns holds one more than the nominal 30). The mask construction is ported verbatim from the reference implementation.

Domain-shift data (independent-source brain MRI). To test how the BraTS-trained prior copes with brain images *unlike* its training data (the central question of this thesis), we use a second, entirely independent dataset: the public *Brain Tumor MRI Dataset* [17], a CC0 aggregation of three sources: the figshare contrast-enhanced-T1 collection [18] ($3{,}064$ slices from 233 patients at two Chinese hospitals), the SARTAJ collection [19], and Br35H [20] (the source of the no-tumour class). It contains $7{,}023$ real clinical brain-MR images in four classes (glioma, meningioma, pituitary, no-tumour). This is a genuine *within-modality distribution shift*: unlike BraTS, these images are *not* skull-stripped (only 19.0% zero pixels), not co-registered or resampled, are stored as lossy JPEG at heterogeneous native resolutions, and mix axial, sagittal and coronal orientations. We centre-crop each image to a square, resize to 240×240, and evaluate on fixed, reproducible samples: 16 images per class (64 total) for the per-class studies and fixed draws of 16–48 evaluation images for the repair studies (stated per experiment), plus disjoint adaptation sets of up to 400 images and 80 held-out images for the adaptation experiments of chapter 7. Because the collection aggregates distinct sources, it also supports the *cross-source* experiment of section 7.4 (adapt on figshare/SARTAJ, evaluate on Br35H). One caveat is inherited from the data: the collection carries no patient identifiers, so counts are of images, not patients, and a patient-wise split is not possible, a limitation we mitigate with cluster-level statistics (section 4.3).

Each of the differences from BraTS matters mechanically, not just cosmetically. The skull is the largest: it is a bright, high-contrast ring occupying the image border, and the prior has never generated one; its score field near the border has only ever pointed toward "black background". JPEG compression introduces 8×8 block artefacts and discards exactly the high-frequency content that undersampled reconstruction must re-invent. Heterogeneous native resolutions mean the effective sharpness varies image-to-image after resizing, and mixed orientations (axial/sagittal/coronal) present silhouettes the co-registered, axially sliced training set under-represents. Chapter 7 disentangles which of these actually cause the failure.

Out-of-distribution data (different modality). For the cross-modality probe we use chest CT from the LIDC dataset [21], resized to 240×240.

Replication and mask data. Two further sets serve the hardening experiments of chapters 6 and 7. *OASIS-1* [22] supplies the second-source replication arm: raw MP-RAGE volumes in native medical format (no lossy compression) with `OAS1` subject identifiers, from which we take one central axial slice per subject, splitting the subject pool into 48 evaluation, 40 adaptation and 20 held-out subjects (disjoint). The *Medical Segmentation Decathlon* brain-tumour task [23] supplies BraTS-preprocessed volumes with expert tumour segmentations, from which we take, for each of 24 patients, the axial slice with the largest tumour area (FLAIR channel). In both sets one slice equals one person, so their counts are patient counts. Table 4.1 summarises.

Table 4.1: Datasets used. N_{img} = number of 2-D images used in this thesis; $H \times W$ = image size.

Dataset	Modality	Role	N_{img}	$H \times W$	Source	Year
BraTS 2021 (train)	Brain MRI	prior training	$297{,}270$ slices	240×240	Synapse/Kaggle	2021
BraTS test (`.npz`)	Brain MRI	evaluation	1000	240×240	released by [1]	2021
Brain Tumor MRI	Brain MRI	domain shift	64 eval $+ 400$ adapt $+ 80$ held-out	var.$\rightarrow 240$	Kaggle/HF (CC0)	2015–21
OASIS-1	Brain MRI	replication	108 subjects (1 slice each)	$256^2 \rightarrow 240^2$	oasis-brains.org	2007
MSD Task01	Brain MRI	real-lesion masks	24 patients (1 slice each)	240×240	Medical Segm. Decathlon	2018
LIDC (CT)	Chest CT	OOD probe	8	$320^2 \rightarrow 240^2$	TCIA	2011

4.2 Evaluation metrics

All quality is measured on real-valued magnitude images clipped to $[0, 1]$ (data range MAX $= 1$). For the in-distribution experiments and the domain-shift reconstructions of chapter 6 the evaluation region is a circular region of interest (ROI) matching the reference protocol; the prior-gain analyses and repair experiments of chapter 7 use the brain foreground mask defined below.

PSNR. For a reconstruction $\hat{\mathbf{x}}$ and ground truth $\mathbf{x}$ of M pixels,

$$\mathrm{MSE}(\hat{\mathbf{x}}, \mathbf{x}) = \frac{1}{M} \sum_{i=1}^{M} (\hat{x}_i - x_i)^2, \qquad \mathrm{PSNR}(\hat{\mathbf{x}}, \mathbf{x}) = 10 \log_{10} \frac{\mathrm{MAX}^2}{\mathrm{MSE}(\hat{\mathbf{x}}, \mathbf{x})}. \tag{4.1}$$

PSNR is logarithmic: $+3$ dB corresponds to halving the mean squared error, $+6$ dB to quartering it.

SSIM. The structural similarity index [24] between local windows of $\hat{\mathbf{x}}$ and $\mathbf{x}$ is

$$\mathrm{SSIM}(\hat{\mathbf{x}}, \mathbf{x}) = \frac{(2\mu_{\hat{x}}\mu_x + C_1)(2\sigma_{\hat{x}x} + C_2)}{(\mu_{\hat{x}}^2 + \mu_x^2 + C_1)(\sigma_{\hat{x}}^2 + \sigma_x^2 + C_2)}, \tag{4.2}$$

where μ, σ^2 and $\sigma_{\hat{x}x}$ are local mean, variance and covariance over an 11×11 Gaussian window ($\sigma = 1.5$), and $C_1 = (0.01L)^2$, $C_2 = (0.03L)^2$ with $L = \mathrm{MAX} = 1$. We report the mean over windows, computed with the `piq` Gaussian-window estimator to match the reference implementation exactly.

Prior-gain: the fair endpoint for domain shift. Absolute PSNR is not a safe endpoint when the experimental arms modify the image statistics (e.g. harmonization, section 7.1): any transformation that simplifies the image (zeroing background, compressing the histogram) inflates the PSNR of *every* reconstructor, prior or not. That would look like an improvement. Our primary endpoint off-domain is therefore the **prior-gain over zero filling, inside the brain**:

$$\Delta \;=\; \mathrm{PSNR}(\hat{\mathbf{x}}_{\mathrm{prior}}, \mathbf{x}) \;-\; \mathrm{PSNR}(\hat{\mathbf{x}}_{\mathrm{zf}}, \mathbf{x}) \;=\; 10 \log_{10} \frac{\mathrm{MSE}(\hat{\mathbf{x}}_{\mathrm{zf}}, \mathbf{x})}{\mathrm{MSE}(\hat{\mathbf{x}}_{\mathrm{prior}}, \mathbf{x})}, \quad \text{over the brain mask,} \tag{4.3}$$

where $\hat{\mathbf{x}}_{\mathrm{zf}} = \mathbf{T}^{-1}\mathbf{\Lambda}\mathbf{T}\mathbf{x}$ is the zero-filled reconstruction from the same measurements. Because Δ is a ratio of MSEs, any factor that rescales *both* errors equally cancels exactly; an "easier target" drops out only insofar as it shifts both arms alike, and a transformation that affects the two arms unequally leaves a residual in Δ, which is why the absolute terms are always printed alongside it. $\Delta > 0$ means the model improves on doing nothing; $\Delta < 0$ means the model is *actively harmful*. The brain mask is computed once from the original image (Otsu threshold [25], fill, erode, largest connected component, dilate) and reused unchanged in every arm, so all arms are scored over an identical region. One honesty note on the name: on skull-bearing images this recipe segments the *head foreground*, skull and scalp included, because the erode–dilate steps shrink the outline by only a few pixels; "in-brain" should therefore be read as "inside the head foreground" throughout. The comparison between arms is unaffected, since every arm uses the identical mask.

In plain words. PSNR asks "how far, on average, is each pixel from the truth?" on a logarithmic scale on which every $+3$ dB means the squared error halved. SSIM asks a more perceptual question, "does the local structure (edges, contrast, texture) look the same?", because two images can have equal pixel-wise error while one looks far more degraded. We report both throughout, and neither is a clinical measure: a reconstruction can score well on both while a diagnostically critical detail is wrong, which is exactly why sections 6.3 and 6.5 probe pathology directly. The prior-gain Δ asks the bluntest question: *is the model better than not using a model?* Unusually for the literature, we will see its answer go negative.

Calibration correlations. For the uncertainty map (section 3.3) we report, per image, the Pearson and Spearman correlations between predicted uncertainty u_i and realised absolute error $e_i = |\bar{x}_i - x_i|$; values near 1 mean the map ranks pixels by error correctly. One caveat on their absolute size: the correlations are computed over the evaluation region, which on BraTS includes the large exactly-zero background, where uncertainty and error

are both trivially small. Part of a high correlation therefore reflects the easy background-versus-tissue separation rather than ranking *within* tissue, so we lean on comparisons of these numbers across matched conditions (in- versus off-distribution, before versus after adaptation, which share the region) rather than on their absolute values.

Risk–coverage curves. For selective prediction (section 7.6), at coverage c we retain the fraction c of in-brain pixels with the *lowest* predicted uncertainty and measure the RMSE over exactly those (the "risk"). Sweeping c traces a risk–coverage curve [26], summarised by its area (AURC; lower is better) [27]. Two reference curves bound any method: deferring pixels at *random* (an uninformative map) and deferring by the *true* error (an unattainable oracle).

4.3 Statistical protocol

The same protocol is used throughout, defined here once.

- **Paired designs.** Wherever two methods can be run on the *same* images from the same random seed, we compare them paired: the quantity analysed is the per-image difference, so image-to-image variability (which is large; per-image PSNR spans > 15 dB off-domain) cancels.
- **Exact Wilcoxon signed-rank tests** [28] for paired comparisons, which assume no normality; at our sample sizes ($n = 24$–48) the exact null distribution is computed, so the smallest attainable p at $n = 24$ is 1.2×10^{-7} (all differences of one sign) and $p = 7 \times 10^{-15}$ at $n = 48$.
- **t-based 95% confidence intervals** $\bar{d} \pm t_{0.975,n-1}\, s/\sqrt{n}$ on the mean paired difference, reported alongside every headline effect.
- **Welch's unequal-variance t-test** [29] for the one unpaired comparison in the thesis (tumour vs no-tumour classes, section 6.3), which compares different images by necessity.
- **Cluster-level guard.** Because the domain-shift collection lacks patient identifiers, slices of one patient could appear as multiple "independent" images. As the most conservative feasible grouping we repeat key tests on the four *class means* ($n = 4$; a signed-rank test at $n = 4$ cannot go below $p = 0.125$, so we use a one-sample t-test there) and report both.
- **Win rates.** Alongside means we report the fraction of images on which the effect has the claimed sign: "improves 100% of images" is a stronger and more interpretable statement than a mean shift.

Multiplicity. The thesis reports on the order of twenty significance tests without a formal family-wise correction, so the p-values must be read in two tiers. The headline effects (reproduction, adaptation, cross-source transfer) have p between 10^{-7} and 10^{-15} and survive any reasonable correction; a Bonferroni adjustment over every test in the thesis leaves them untouched. The handful of results with p in the 0.02–0.05 range (the tumour-fidelity Welch comparisons of section 6.3 and the residual 0.71 dB support-constraint cost of section 7.2) would not all survive such a correction and are therefore to be read as *exploratory*: suggestive, consistent with the surrounding evidence, but not confirmatory on

their own. A middle band exists as well: the $4\times$ and $24\times$ in-lesion error elevations of table 6.4 ($p \approx 0.002$–0.007) do not individually survive a Bonferroni adjustment over all twenty tests, but the $8\times$ elevation does ($p_{\text{adj}} \approx 0.002$), and the three are mutually consistent, so we read them jointly. No conclusion of the thesis rests on the exploratory tier alone.

In plain words. Every comparison in this thesis is, wherever physically possible, "the same scan reconstructed both ways": each scan is its own control, so differences between scans cannot masquerade as differences between methods. Significance is computed with tests that do not assume the data are bell-shaped, and every headline number carries both a confidence interval and the fraction of individual images that agree with it. Where a p-value is only modest, the text says so.

5. IN-DISTRIBUTION RESULTS

All experiments use the converted pretrained prior (section 3.1) with the paper's tuned hyperparameters ($\eta = 0.577$, $\lambda = 0.982$), each on a single GPU: a 24 GB RTX 4090 for the reproduction and solver-ablation runs of section 5.1, and an NVIDIA A100 for the multi-sample, adaptive-acquisition and speed experiments. PSNR and SSIM are as defined in eqs. (4.1) and (4.2) (circular ROI, data range MAX $= 1$, `piq` Gaussian-window SSIM). This chapter evaluates the method on its home ground, the BraTS test data, drawn from the same distribution the prior was trained on; chapter 6 then leaves that home ground.

5.1 Reproduction of the published results

The first question is whether the reimplementation is *correct*, in the only sense that can be verified externally: does it reproduce the published numbers? Table 5.1 compares our default solver (`projection`: PC sampling + data consistency, one sample per scan) against the values reported by [1] and against the zero-filled reconstruction $\mathbf{T}^{-1}\mathbf{\Lambda T x}$, over $N_{\text{sl}} = 256$ test slices.

Table 5.1: Undersampled-MRI reproduction on BraTS ($N_{\text{sl}} = 256$ slices). "Ours" is our PyTorch `projection` solver; "Paper" is Table 1 of [1]; "Zero-filled" is the inverse FFT of the masked k-space. Each cell is PSNR (dB) / SSIM; R is the acceleration factor (fraction $1/R$ of k-space kept).

R	Ours (PSNR/SSIM)	Paper (PSNR/SSIM)	Zero-filled (PSNR/SSIM)
$4\times$	40.03 / 0.967	39.91 / 0.965	28.64 / 0.759
$8\times$	37.75 / 0.960	37.63 / 0.958	26.28 / 0.748
$24\times$	29.57 / 0.881	29.42 / 0.880	20.79 / 0.515

Our reimplementation matches the paper to within 0.15 dB PSNR and 0.002 SSIM at every acceleration, agreement well inside run-to-run variation. This validates the reimplementation, the weight conversion, and the measurement pipeline simultaneously (an error in any of the three would appear as a systematic gap). The margin over zero filling is large on this in-distribution data: $+11.4$, $+11.5$ and $+8.8$ dB at $4/8/24\times$. Figure 5.1 shows example $8\times$ reconstructions: the aliasing and blur of the zero-filled input are removed and fine anatomy is restored.

As a sanity check we ran the three alternative solver modes at $8\times$ on $N_{\text{sl}} = 64$ slices: Score-SDE baseline 35.15 / 0.946; Langevin baseline 35.05 / 0.946; ALD+data-consistency 36.72 / 0.951; for reference, PC+data-consistency is the $N_{\text{sl}} = 256$ value of table 5.1, 37.75 / 0.960. The two data-consistency variants dominate the two baselines, reproducing the paper's qualitative ordering. The one discrepancy is the plain Langevin baseline, which under-performs the paper's reported value (35.05 vs 36.44). We attribute this to the released configuration's baseline-specific hyperparameters and note that it does not affect the contribution method, but we report it rather than omit it.

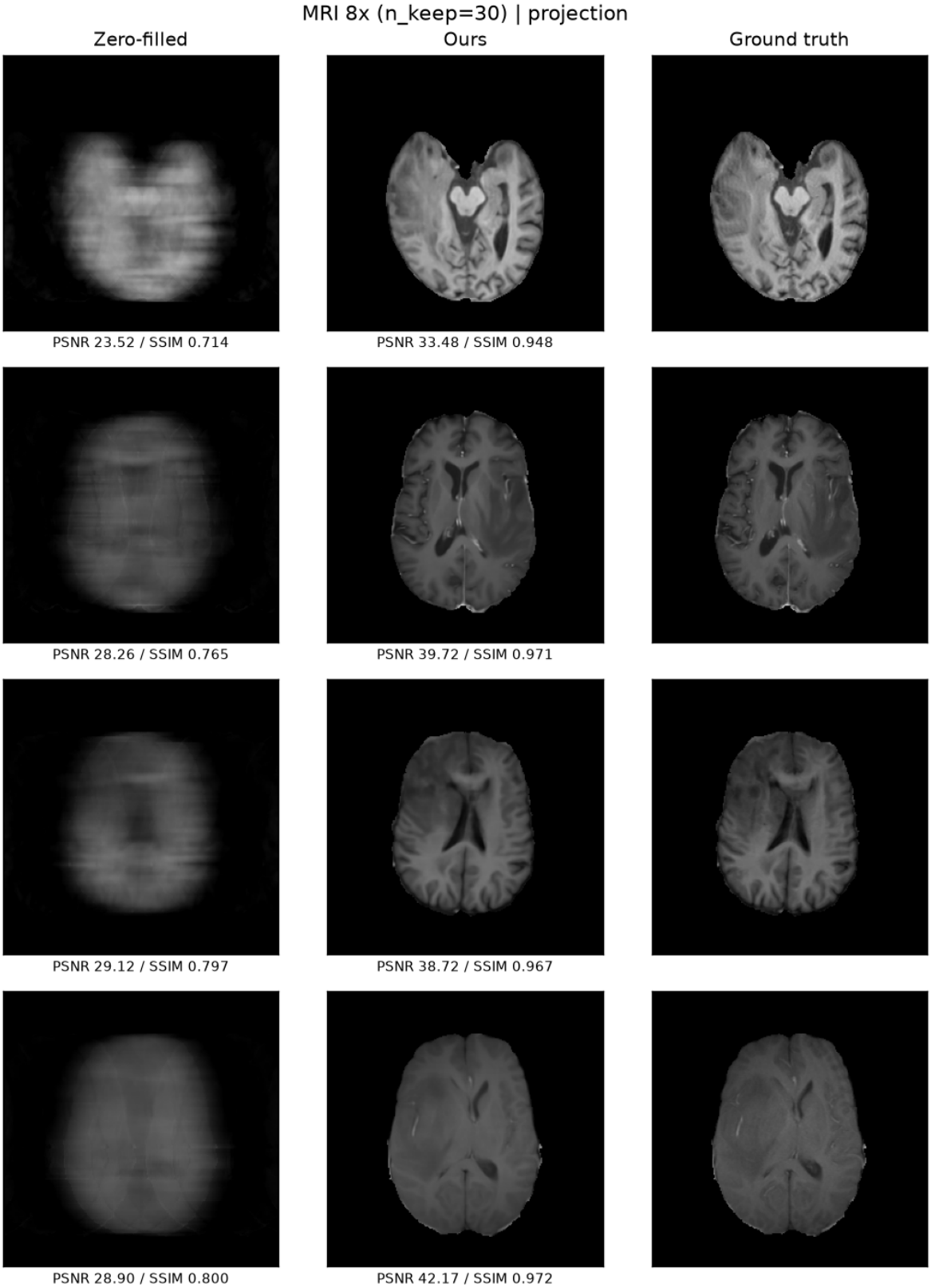


Figure 5.1: Example $8\times$ reconstructions. Columns: zero-filled input, our reconstruction, ground truth. Per-slice PSNR/SSIM are annotated.

5.2 Uncertainty-guided reconstruction

Table 5.2 reports the Monte-Carlo posterior mean (eq. (3.3), $S = 20$ samples) over $N_{\text{sl}} = 64$ slices, against a single posterior sample and against the paper's single-run numbers, to-

gether with the error–uncertainty calibration. Note the protocol difference from table 5.1: these are 64 slices with 20 reconstructions each, so the two tables are not directly comparable cell-by-cell.

Table 5.2: Uncertainty-guided reconstruction ($S = 20$ samples, $N_{\text{sl}} = 64$ slices). "Mean" is the MMSE estimate $\bar{\mathbf{x}}$; "Single" is one posterior sample; "Paper" is the single-run value of [1]. The last column is the Spearman / Pearson correlation between the uncertainty map $\mathbf{u}$ and the true error (section 4.2).

R	Mean	Single	Paper	$\mathbf{u} \leftrightarrow$err (Sp./Pe.)
$4\times$	42.17 / 0.979	40.14 / 0.968	39.91 / 0.965	0.81 / 0.67
$8\times$	39.84 / 0.972	37.93 / 0.961	37.63 / 0.958	0.84 / 0.69
$24\times$	30.95 / 0.903	29.51 / 0.883	29.42 / 0.880	0.87 / 0.72

Three observations. First, averaging the $S = 20$ samples improves PSNR by 1.4–2.0 dB over a single sample at every acceleration, thereby *exceeding* the paper's single-run numbers by 1.5–2.3 dB. This is the MMSE effect anticipated in section 2.6, and it comes at $20\times$ the reconstruction cost, a trade-off we state plainly; the speed result of section 5.4 halves that cost. The gain also follows the perception–distortion behaviour of section 2.6: clinically, a radiologist preferring sharper texture might still prefer a single sample. Second, a single sample from our solver already matches the paper, so the improvement comes from aggregation rather than from a better base solver. Third, the uncertainty map ranks pixels by their true error well (Spearman 0.81–0.87), and the ranking *strengthens* with acceleration: at $24\times$ more of the image is inferred rather than measured, and the map tracks exactly that. Figure 5.2 shows the map concentrating on anatomical boundaries and inferred structure.

Choice of aggregator. Replacing the mean by the per-pixel median or by a middle-$S/2$ trimmed mean changes the result negligibly (table 5.3; $N_{\text{sl}} = 8$): all three agree within 0.1 dB, because the data-consistency filter of section 3.3 has already removed outlier samples. The mean is marginally best at low acceleration; the robust aggregators are marginally better in SSIM at $24\times$. The choice of aggregator is therefore immaterial in this regime, and we use the mean throughout.

Table 5.3: Aggregator comparison ($S = 20$ samples, $N_{\text{sl}} = 8$ slices). Cells are PSNR / SSIM.

R	Mean	Median	Trimmed (mid-$S/2$)
$4\times$	42.46 / 0.980	42.37 / 0.980	42.42 / 0.980
$8\times$	40.16 / 0.974	40.07 / 0.974	40.13 / 0.974
$24\times$	31.46 / 0.910	31.45 / 0.915	31.50 / 0.915

5.3 Uncertainty-guided adaptive acquisition: a negative result

Table 5.4 tests whether the uncertainty can select *which* $B = 20$ extra k-space columns to acquire (section 3.4), starting from the $8\times$ mask (nominally $n_{\text{keep}} = 30$ columns; the equispaced construction always includes the central column, so 31 are actually sampled) and reaching $4.7\times$ (51 columns), over $N_{\text{sl}} = 32$ slices with $K = 8$ posterior samples for the importance map eq. (3.4).

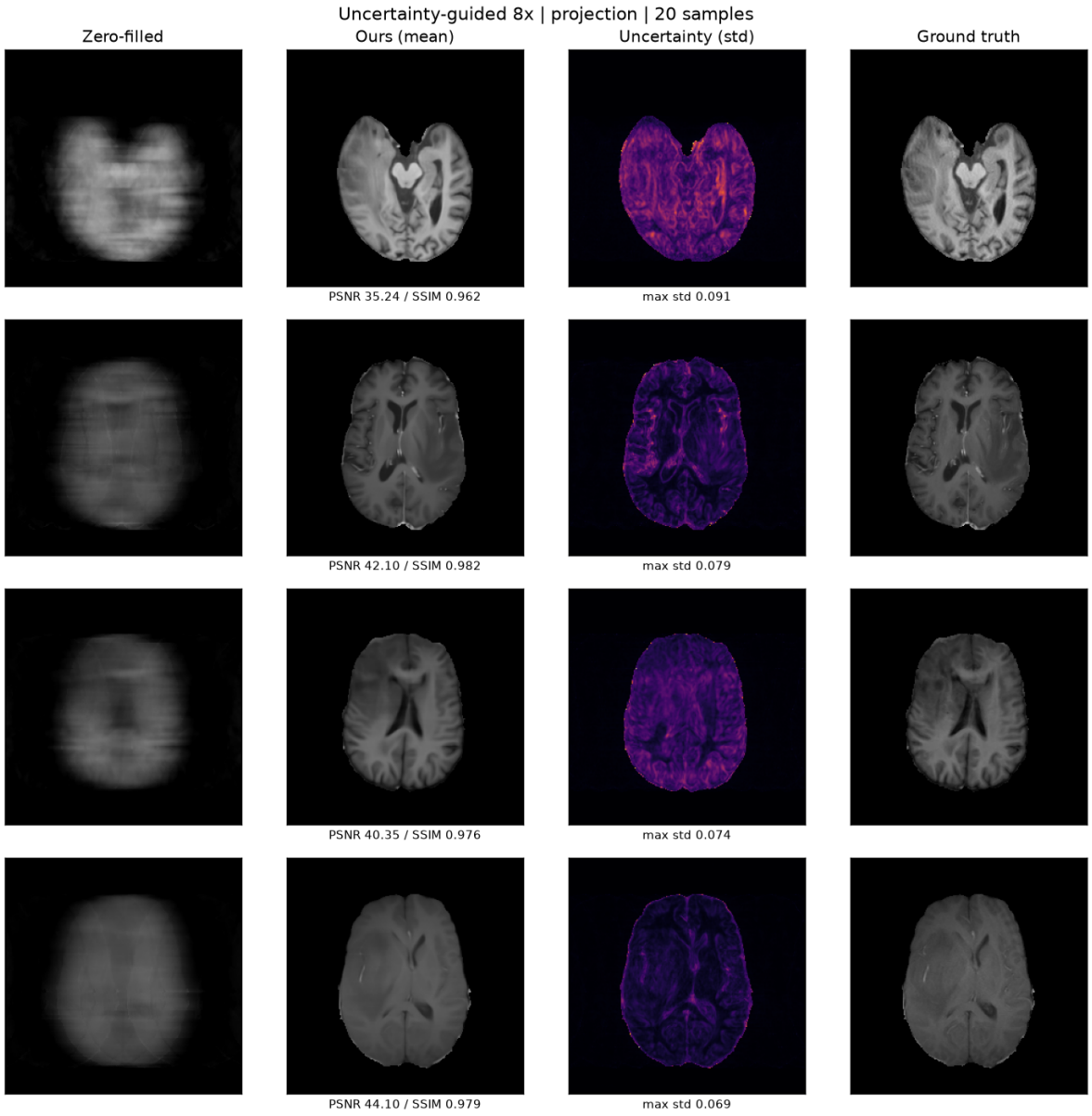


Figure 5.2: Uncertainty-guided reconstruction at $8\times$. Columns: zero-filled input, MMSE mean, per-pixel uncertainty (brighter $=$ higher **u**), ground truth.

Table 5.4: Adaptive acquisition: $+20$ columns by three selection strategies (same budget), built from $K = 8$ posterior samples, over $N_{\text{sl}} = 32$ slices. "Paired vs random" is the mean per-slice PSNR difference against `random`, its standard error, and the fraction of slices on which the strategy wins.

Selection ($+20$ cols $\rightarrow 4.7\times$)	PSNR	SSIM	Paired vs random
initial ($8\times$, 31 cols)	39.69	0.972	—
guided (uncertainty)	43.69	0.984	-0.06 ± 0.21 dB, wins 31%
random	43.75	0.984	—
low-frequency	42.59	0.981	-1.16 ± 0.15 dB, wins 9%

Under the paired per-slice test, uncertainty-guided selection is *not* better than random: -0.06 dB (mean $\pm$ SEM -0.06 ± 0.21; t-based 95% CI $[-0.48, +0.36]$, n=32), winning only 31% of slices, which amounts to a statistical tie. A preliminary run on $N_{\text{sl}} = 4$ slices had

suggested a $+0.48$ dB advantage, which evaporated under the properly powered paired analysis. Small margins evaluated on small samples are noise until proven otherwise; this experiment is the thesis's concrete demonstration of that principle. The only robust effect is that guided and random alike beat the naive `low-frequency` heuristic (by $+1.10$ and $+1.16$ dB respectively), which merely crowds samples where the mask is already dense. We treat the null as a result in its own right.

Why does an intuitively sensible idea fail? We can offer only labelled speculation: with a Cartesian mask the choice is between whole *columns*, a coarse unit that may average away whatever per-region signal the uncertainty carries; and once the low frequencies are covered, the marginal value of any single high-frequency column may simply be too uniform for *any* ranking to beat a random draw. Distinguishing these explanations would require per-scan (rather than batch-shared) selection and finer acquisition units; both are listed as future work in chapter 9. What the experiment does establish, without speculation, is the value of its own design: the paired test with an adequate number of slices is what separated a publishable-looking $+0.48$ dB from the truth.

5.4 Sampling speed

Table 5.5 varies the number of reverse-diffusion steps N at $8\times$; each step costs a fixed number of score evaluations, so compute is proportional to N. Because every 500-step protocol downstream rests on this ablation, it was re-measured on $N_{\text{sl}} = 32$ slices (an earlier $N_{\text{sl}} = 8$ pilot gave the same ordering).

Table 5.5: Quality vs. number of reverse-diffusion steps N ($8\times$, $N_{\text{sl}} = 32$ slices, single sample). Δ is the PSNR change relative to the $N{=}1000$ baseline.

Steps N	1000	500	200	100	50
PSNR	37.89	37.44	36.19	34.61	32.75
SSIM	0.961	0.958	0.950	0.936	0.912
Δ PSNR	—	-0.46	-1.71	-3.29	-5.14

Halving the steps to $N = 500$ costs 0.46 dB, essentially full quality at half the compute. This in turn halves the score-evaluation budget of the 20-sample uncertainty method of section 5.2, and is the setting adopted for the large re-measurement of section 7.4. In concrete terms, a single default reconstruction costs $\approx$2000 score evaluations of the 2.7M-parameter network; the 20-sample MMSE method costs $\approx$40,000; at $N = 500$ this falls to $\approx$20,000 for the same final quality. That is the difference between the two protocols used in section 7.3 and section 7.4. Figure 5.3 plots the full ablation of table 5.5.

$N = 200$ (-1.7 dB) remains usable when speed dominates. Replacing the stochastic predictor with the deterministic probability-flow ODE (eq. (2.15)) at equal step count gives the same quality (in the $N_{\text{sl}} = 8$ pilot: $36.46\,/\,0.950$ vs $36.46\,/\,0.952$ at $N = 200$): for this solver, determinism buys nothing at fixed N. (Wall-clock does not fall exactly $2\times$ at $N = 500$ because of the fixed per-run overhead discussed in section 3.5; hence the effect is stated in score evaluations, the quantity that actually scales.)

5.5 Measurement noise

The forward model used so far is noise-free, which flatters any method. To measure the cost of realism we add complex Gaussian noise to the measured k-space lines, with the

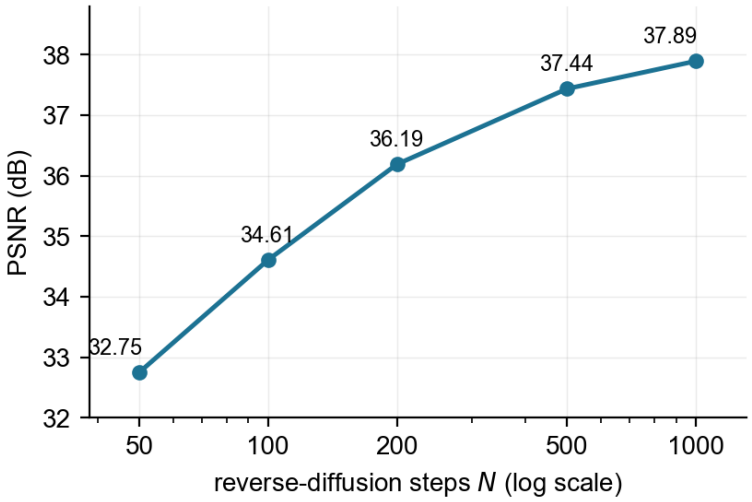


Figure 5.3: Reconstruction quality against the number of reverse-diffusion steps N ($8\times$, $N_{\text{sl}} = 32$ slices, single sample; the data of table 5.5). The knee sits at $N = 500$: half the compute for 0.46 dB.

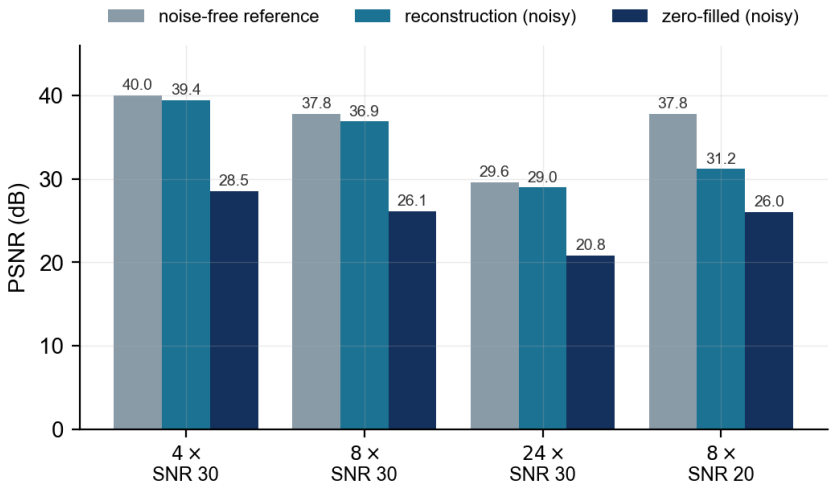


Figure 5.4: Reconstruction under complex Gaussian k-space noise (the data of table 5.6). Bars show the noise-free reference, the reconstruction from noisy measurements, and the noisy zero-filled baseline.

per-image noise power set from the measured-line signal power at a target signal-to-noise ratio: $\sigma^2 = \langle |\mathbf{Tx}|^2 \rangle_{\text{measured}} / 10^{\text{SNR}/10}$, split evenly between the real and imaginary parts. The solver sees exactly what a scanner would hand it (the data-consistency merge uses the noisy measurement); PSNR is still computed against the clean image. Table 5.6 repeats the single-sample protocol of table 5.1 on $N_{\text{sl}} = 64$ slices; fig. 5.4 plots the same data.

Table 5.6: Reconstruction under complex Gaussian k-space noise ($N_{\text{sl}} = 64$ slices, single sample, 1000 steps). Cells are PSNR(dB) / SSIM (circular ROI); "zero-filled" is the noisy zero-filled baseline.

Setting	Reconstruction	Zero-filled	Noise-free reference
$4\times$, SNR 30 dB	39.38 / 0.952	28.51 / 0.749	40.03 / 0.967
$8\times$, SNR 30 dB	36.87 / 0.941	26.12 / 0.736	37.75 / 0.960
$24\times$, SNR 30 dB	28.96 / 0.865	20.82 / 0.509	29.57 / 0.881
$8\times$, SNR 20 dB	31.17 / 0.792	26.00 / 0.678	37.75 / 0.960

At a realistic SNR of 30 dB the method loses 0.61–0.88 dB relative to the noise-free forward model and its margin over zero filling is essentially unchanged. At a harsher 20 dB the loss grows to 6.6 dB at $8\times$, yet the reconstruction still leads the zero-filled baseline by 5.2 dB. The noise-free headline numbers are therefore an idealisation, but not a fragile one. The same holds off-domain: repeating one domain-shift configuration ($8\times$, SNR 30 dB, $N_{\text{img}} = 32$ Brain Tumor MRI images, $S = 4$) gives a prior-gain of $+0.86$ dB $[-1.32, +3.04]$ (59% wins), statistically indistinguishable from zero and consistent with the noiseless finding that the mismatched prior adds no reliable value at $8\times$ (section 6.2).

6. FAILURE MODES UNDER DOMAIN SHIFT

A reconstruction prior that is to be trusted clinically must be characterised by *where it breaks*, not just by its average accuracy on curated data. Three questions structure this chapter. How does the method behave on content it was never trained on, whether a different modality altogether (section 6.1) or, more realistically, the same modality acquired differently (section 6.2)? Does it treat *pathology*, the reason the scan exists, worse than healthy tissue (section 6.3)? And at what combination of feature size and acceleration does it silently *erase* a finding (section 6.5)? The chapter closes by stating the method's scope (section 6.6); chapter 7 then asks *why* the failures happen and what repairs them.

6.1 Distribution specificity: a different modality

The prior was trained only on 240×240 skull-stripped brain-MRI slices (section 4.1); nothing constrains its behaviour elsewhere. As the bluntest probe, we reconstruct chest-CT slices (LIDC, resized to 240×240) through the *same* brain-MRI prior and $8\times$ mask, and compare against the in-distribution brain-MRI baseline, both with $S{=}20$ posterior samples over 8 slices (table 6.1).

Table 6.1: In-distribution vs. out-of-distribution reconstruction ($8\times$, $S{=}20$ samples, 8 slices). "$\mathbf{u} \leftrightarrow$err" is the Spearman correlation of section 4.2. The in-distribution row is computed under this experiment's evaluation pipeline, which min–max rescales each slice to $[0, 1]$ (required to bring CT onto the prior's intensity scale); this stretches the BraTS slices and lowers PSNR by ≈ 2.5 dB relative to the protocol of table 5.2, so the row is not comparable cell-by-cell with chapter 5; the in- vs out-of-distribution comparison *within* the table is unaffected.

Input	PSNR	SSIM	$\mathbf{u} \leftrightarrow$err (Sp.)
Brain MRI (in-distribution)	37.61	0.967	0.837
Chest CT (out-of-distribution)	19.38	0.580	0.423

The reconstruction collapses: PSNR falls from 37.61 to 19.38 dB and SSIM from 0.967 to 0.580, because the brain-MRI prior imposes brain-like structure on content it never saw. The data-consistency step still forces agreement with the measured chest-CT k-space, but on the *unmeasured* coordinates the sampler follows a score field trained exclusively on brains, so the missing seven-eighths of the spectrum are filled with the wrong anatomy's statistics. Out of distribution, the uncertainty is *also* much less reliable: its rank correlation with the true error drops from 0.837 to 0.423. This is the more dangerous half of the result. The uncertainty map measures the *posterior spread* (how much the model's answers disagree with each other), not their distance from the truth; a prior that is confidently wrong produces samples that agree on the wrong answer, and the map goes quiet exactly where the reconstruction is most wrong. That makes it least trustworthy in the very cases that call for it, and the immediate lesson is that both the reconstruction *and its safety signal* must be re-validated before use on any new modality, anatomy, or scanner.

6.2 Independent-source brain MRI: the realistic case

The CT experiment changes modality *and* anatomy at once, an extreme case no one would deploy. The realistic hazard is subtler: the *same* anatomy and modality, acquired dif-

ferently. We therefore reconstruct the independent Brain Tumor MRI dataset (section 4.1): real clinical brain MRI that, unlike BraTS, is not skull-stripped, not co-registered or resampled, JPEG-compressed at heterogeneous resolutions, and mixed in orientation. Table 6.2 reports the MMSE reconstruction ($S{=}8$ samples) per class at $4\times$, $8\times$ and $24\times$ over 16 images per class.

Table 6.2: Reconstruction of independent-source brain MRI by the BraTS-trained prior (MMSE mean, $S{=}8$ samples, 16 images per class). Cells are PSNR(dB) / SSIM (circular ROI, data range 1).

Class	$4\times$ (PSNR/SSIM)	$8\times$ (PSNR/SSIM)	$24\times$ (PSNR/SSIM)
glioma	27.74 / 0.829	21.55 / 0.696	14.79 / 0.520
meningioma	26.36 / 0.821	21.03 / 0.699	14.40 / 0.506
no-tumour	23.13 / 0.722	16.24 / 0.558	12.19 / 0.416
pituitary	23.07 / 0.738	17.22 / 0.573	11.92 / 0.420

Reading the table: each cell is the average over 16 images of that class, reconstructed with the full uncertainty method (mean of $S = 8$ posterior samples); columns move from mild ($4\times$) to extreme ($24\times$) undersampling. For scale, recall that decibels are logarithmic: a 14 dB drop means the squared error grew by a factor of ~ 25.

Two things stand out. First, fidelity drops sharply relative to in-distribution BraTS (MMSE $42.2/39.8/30.9$ dB at $4/8/24\times$, table 5.2): here even the easiest class reaches only 27.7 dB at $4\times$ (~ 14 dB lower) and 14.8 dB at $24\times$. The MMSE mean still improves on a single sample (by 1.1–2.0 dB at $4\times$, shrinking to 0.4–0.8 dB at $24\times$), so the uncertainty averaging keeps helping off-distribution, though less at extreme acceleration. Second, and more seriously, the prior's benefit degrades *unevenly, then universally*. At $4\times$ every class still beats zero filling, but by uneven margins: $+4.6/+3.8$ dB for the tumour classes (figshare/SARTAJ) against $+2.8$ dB for no-tumour and only $+0.7$ dB for pituitary. At $8\times$ the split becomes qualitative: glioma and meningioma remain (barely) above zero filling (21.55 vs 20.99, 21.03 vs 20.19 dB) while the less-BraTS-like no-tumour (Br35H) and pituitary classes fall *below* it (16.24 vs 18.66, 17.22 vs 20.05 dB). At $24\times$ *every* class falls below the zero-filled baseline (glioma 14.8 vs 18.1; pituitary worst, 11.9 vs 17.5 dB): the mismatched prior imposes wrong structure and *actively degrades* the image, at extreme acceleration even for the classes it handles best. Calibration weakens correspondingly (Spearman 0.31–0.67 across classes, versus 0.84 in-distribution) and is worst on the classes the prior handles worst. A model that can be worse than no model, while its own reliability signal degrades, is the central safety fact this thesis establishes; quantifying and repairing it occupies chapter 7. Figure 6.1 shows a representative glioma reconstruction.

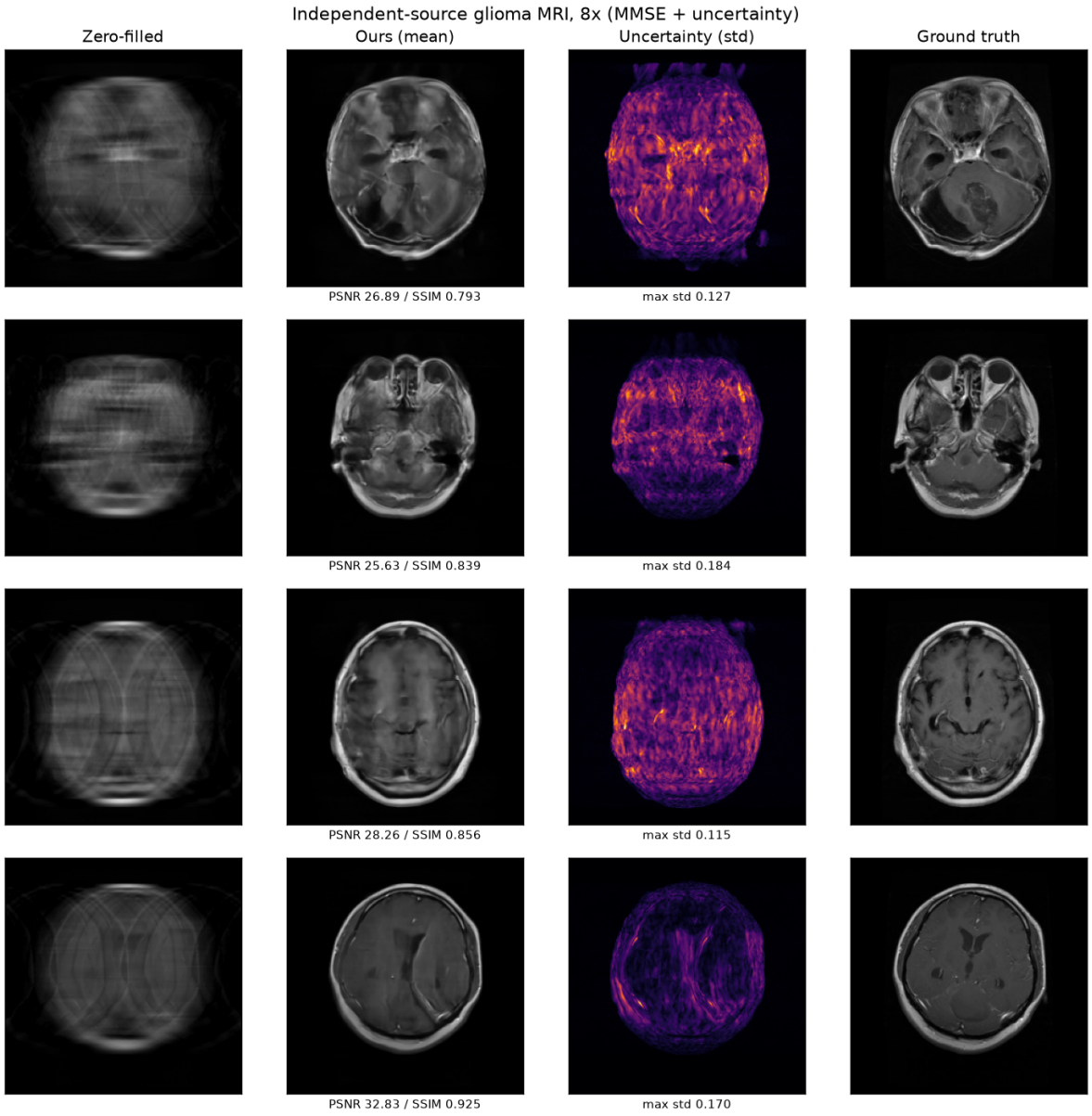


Figure 6.1: Independent-source glioma MRI reconstructed at $8\times$ by the BraTS-trained prior. Columns: zero-filled input, MMSE mean, per-pixel uncertainty, ground truth. Note the skull, absent from the training data. Per-slice PSNR varies widely; the class average is table 6.2's 21.55 dB.

6.3 Real-tumour fidelity

The clinically sharpest worry about a generative prior is that it might selectively *erase* the very pathology one is scanning for, reconstructing the typical at the expense of the atypical. Because three of the four classes above carry a *real* tumour and one is tumour-free, this dataset lets us test the worry without synthetic proxies: does the prior reconstruct genuinely pathological slices worse than healthy ones? Table 6.3 pools the three tumour classes and compares their per-image PSNR against the no-tumour class with a Welch test (unpaired by necessity, since they are different images); section 6.4 then removes the between-source confound with a within-image design under expert tumour masks.

Table 6.3: Real-pathology fidelity: pooled tumour classes (glioma+meningioma+pituitary, $n{=}48$) vs. no-tumour ($n{=}16$), per-image PSNR of the MMSE reconstruction. Δ is tumour$-$no-tumour; p from Welch's test.

R	Δ (dB)	Welch p
$4\times$	$+2.60$	0.230
$8\times$	$+3.70$	0.038
$24\times$	$+1.52$	0.041

Contrary to the worst-case worry, tumour-bearing slices are reconstructed no worse than healthy ones; they are in fact better, by 1.5–3.7 dB, and significantly so at $8\times$ and $24\times$. We read this cautiously: the tumour and no-tumour images also come from *different source collections* (figshare/SARTAJ vs Br35H), so the gap reflects acquisition and image quality as much as pathology per se, a confound we cannot remove with this artefact. The defensible conclusion is negative-but-reassuring: on this data there is *no evidence of systematic tumour erasure* at 4–$24\times$. Erasure *does* occur under heavy undersampling, and the smaller the feature the further it goes, which the controlled experiment below isolates.

6.4 Real tumours under expert masks: within-image fidelity

The comparison above is between *different images* from different sources, so acquisition differences confound it (a limitation section 4.3 flags as exploratory). A sharper design asks the question *within* a single image: does the reconstruction err more inside a real tumour than in the surrounding healthy tissue of the same slice? We use the Medical Segmentation Decathlon brain-tumour task (BraTS-preprocessed volumes with expert tumour segmentations), taking for each of 24 patients the axial slice with the largest tumour area (FLAIR channel, one slice per patient). Because these volumes share BraTS preprocessing, no pipeline shift enters; because the comparison is in-slice, no acquisition confound enters. For each patient we compare the mean absolute error and the mean uncertainty inside the expert mask against the brain tissue outside it, paired across patients (exact Wilcoxon).

Table 6.4: Within-image fidelity at real tumours ($N_{\text{img}} = 24$ patients, one slice each, $S{=}6$ samples). "in/out" is the ratio of the in-lesion to out-of-lesion value, averaged over patients; p from the exact Wilcoxon test on the paired in-vs-out difference.

Acceleration	Error in/out	p	Uncertainty in/out	p
$4\times$	$1.10\times$	0.007	$1.11\times$	0.002
$8\times$	$1.22\times$	1.1×10^{-4}	$1.21\times$	1.3×10^{-4}
$24\times$	$1.37\times$	0.005	$1.26\times$	0.065

Figure 6.2 shows the paired comparison per patient. Two findings emerge. First, tumour tissue *is* reconstructed worse than the healthy tissue of the same slice: the in-lesion error exceeds the out-of-lesion error by 10% at $4\times$, 22% at $8\times$ and 37% at $24\times$, all significant under the paired test. The effect is modest at clinical accelerations and grows with the fraction of the image the prior must invent. This sharpens, rather than contradicts, the between-image comparison above: pathology is not selectively *erased*, but it is the hardest content in the slice, and the reconstruction is systematically least faithful there. Second,

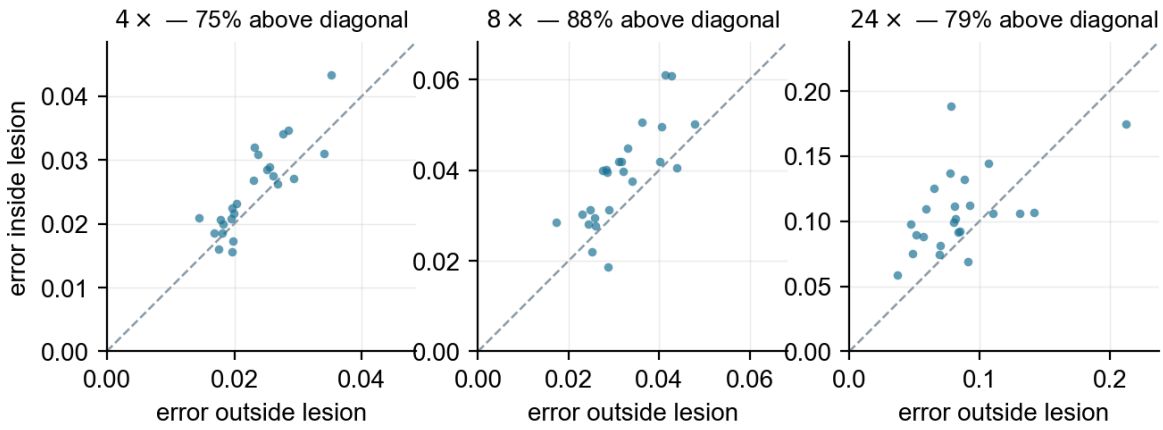


Figure 6.2: Per-patient mean absolute error inside the expert tumour mask against the error in the surrounding brain tissue of the same slice ($N_{\text{img}} = 24$ patients). Points above the diagonal have higher in-lesion error; the fraction above the diagonal is $75/88/79\%$ at $4/8/24\times$.

the uncertainty map tracks this: in-lesion uncertainty is elevated by almost the same factor as the in-lesion error at $4\times$ and $8\times$ ($p \leq 0.002$), so the map points a reader at the right tissue; at $24\times$ the elevation ($1.26\times$) no longer reaches significance ($p = 0.065$), consistent with section 6.2: the safety signal weakens as the risk it is meant to flag grows.

6.5 Rare features: the erasure threshold

Real tumours in section 6.3 are large. The remaining hazard is the *small, rare* finding [30–32]. We probe it under full control: into each BraTS slice we insert one bright Gaussian "lesion" of amplitude c and radius ρ pixels (the Gaussian's standard deviation; σ stays reserved for the VE noise scale), treat the result as ground truth, undersample, reconstruct, and measure (i) the fraction of the inserted contrast that survives, and (ii) the error and uncertainty inside vs outside the lesion. Since BraTS is 1 mm isotropic, 1 pixel $= 1$ mm; the visible core of a Gaussian spot is roughly its full width at half maximum, 2.355ρ, so $\rho = 8$ px is a $\approx$19 mm feature (a sizeable mass) and $\rho = 4$ px a $\approx$9 mm one (a small metastasis).

At the baseline configuration (c=0.5, ρ=8 px, $8\times$, S=20 samples, 8 slices) the lesion survives essentially intact: contrast recovered $0.334/0.335 \approx 100\%$. The reconstruction is measurably worse at the lesion (absolute error 0.0310 vs 0.0161 outside, $1.9\times$) and, encouragingly, the uncertainty is elevated exactly there (0.0181 vs 0.0131, $1.38\times$): the map flags the region it reconstructs least well. This is reassuring but configuration-specific, so table 6.5 sweeps size and acceleration to find where survival stops.

Table 6.5: Lesion-erasure sweep (added contrast c=0.5, 8 slices, S=6 samples). "Recovered" is the reconstructed lesion contrast as a fraction of the inserted contrast (near 1 = survives, $\ll 1$ = erased), with the across-slice standard deviation; "err in/out" is the absolute error inside vs outside; "**u**-ratio" the uncertainty inside over outside.

Size (ρ, px = mm)	Acceleration	Recovered	Err in / out	**u**-ratio
8 (≈19 mm)	$8\times$	$98 \pm 5\%$	0.021 / 0.016	$1.48\times$
4 (≈9 mm)	$8\times$	$95 \pm 5\%$	0.022 / 0.015	$1.55\times$
8 (≈19 mm)	$24\times$	$33 \pm 35\%$	0.182 / 0.046	$1.89\times$
4 (≈9 mm)	$24\times$	$2 \pm 59\%$	0.241 / 0.041	$1.84\times$

The sweep brackets the failure. At $8\times$ a strong lesion is essentially preserved (95–98%) regardless of size, with small across-slice spread. At $24\times$ it is largely *erased*: on average 33% of a 19 mm feature survives and $\sim$2% of a 9 mm one, though the across-slice spread is large at this acceleration (± 35 and ± 59 percentage points at $n = 8$), so the two recovered fractions are not individually separable; the trend toward more complete destruction of the smaller feature is consistent with its larger in-lesion error but is not resolved on its own. Meanwhile the in-lesion error rises to 4–$6\times$ the background and the uncertainty rises with it ($\sim 1.85\times$), so the map flags the erased region. For fainter insertions ($c{=}0.25, 0.1$) the contrast-recovery ratio is ill-defined (the spot barely exceeds the surrounding intensity, so its denominator collapses) and is omitted; there the diagnostic is the in/out error and uncertainty ratio, which at $8\times$ are close to 1 (**u**-ratio 1.03–1.28): a faint lesion blends into the reconstruction and is *not* flagged. The clinical caution is concrete: at aggressive acceleration, or for low-contrast findings, rare features can be silently lost; the uncertainty map catches the strongly erased cases but not the faint ones.

6.6 Scope and other limitations

Beyond the failure modes measured above, four boundaries define what this thesis does and does not claim.

- **Retrospective simulation.** Undersampling is simulated from existing 2-D magnitude images with a single-coil forward model, matching [1]. The headline results use a noise-free model; section 5.5 measures the cost of realistic k-space noise (≤ 0.9 dB at SNR 30 dB). There is no prospective scanning, no multi-coil or complex-valued data, no reader study, and no clinical validation; PSNR/SSIM are surrogate metrics.
- **Counts are images, not patients, except where stated.** The Kaggle collection carries no patient identifiers, so several of its slices may derive from one person; the cluster-level tests of section 4.3 are the guard there. The OASIS-1 replication (section 7.5) and the expert-mask study (section 6.4) use one slice per person, so their counts *are* patient counts.
- **The uncertainty map is a ranking, not a guarantee.** This was established in section 5.2 and is bounded by the faint-lesion result above.
- **Temporal shift is sampled, not covered.** The prior encodes BraTS as of 2021; protocol shift is measured through two independent collections (section 6.2, acquired 2015–21; section 7.5, 2007). Neither post-dates the training corpus, so genuinely newer acquisition protocols remain untested.

7. DIAGNOSING AND REPAIRING THE DOMAIN-SHIFT FAILURE

Chapter 6 established *that* the prior fails off-source. This chapter establishes *why*, and what fixes it. The argument proceeds in four controlled steps: an oracle harmonization study isolates the cause (section 7.1); a deployable version of the implied remedy (constraining the input) turns out neutral at best, after a control exposes a bug in our own first implementation (section 7.2); adapting the *model* instead works decisively (sections 7.3 and 7.4); and the calibrated uncertainty becomes an actionable decision rule (section 7.6).

7.1 Diagnosis: an oracle harmonization study

Is the failure of section 6.2 caused by the *input statistics*, or is the prior simply useless off-domain? To find out we apply a three-step harmonization to each image, leaving the model and sampler untouched: (1) a crude skull-strip (Otsu threshold [25] $\rightarrow$ fill $\rightarrow$ erode $\rightarrow$ largest connected component $\rightarrow$ dilate), because BraTS is skull-stripped (73.2% exactly-zero pixels vs 19.0% here), and a pretrained extractor such as SynthStrip [33] would do this properly; (2) percentile normalization (clip to $[p_1, p_{99}]$ of the foreground, rescale to $[0, 1]$), reproducing the training data's robust per-volume scaling (scale matters because σ_{max} of the VE-SDE is calibrated to the training intensity range [34]); (3) Nyúl-style histogram matching [35] of the foreground to a reference CDF pooled from 200 real BraTS slices.

This is an oracle: a diagnosis, not a remedy. The harmonization is applied to the image *before* k-space is simulated: we compute $\mathbf{x}' = \text{harmonize}(\mathbf{x})$ and then measure $\mathbf{y} = \mathbf{A}\mathbf{x}'$. A real scanner cannot do this: the skull is physically in the patient, its energy is in the acquired k-space, and removing it beforehand would require the fully sampled image one is trying to avoid acquiring. Every number in this section is an *oracle upper bound* that identifies what the prior needs, not a deployable preprocessing step. The deployable versions are sections 7.2 and 7.3.

A fair endpoint. Harmonization also zeroes the background and smooths the histogram (Shannon entropy falls from 4.32 to 3.44–3.49 bits), which inflates *absolute* PSNR for any reconstructor whatsoever. Our endpoint is therefore the prior-gain Δ of eq. (4.3) (reconstruction minus zero filling, inside the same arm, over the brain only), from which any effect common to both arms cancels; arm-specific effects do not, and the decomposition below exhibits one (contrast stretching penalises the zero-filled arm more than the reconstruction), which is why both absolute terms are printed and `mask`, whose zero-fill baseline barely moves, is treated as the headline. Table 7.1 reports Δ over $N_{\text{img}} = 24$ images at $8\times$, each arm compared to `native` *paired* on the same images, plus a pre-specified control arm.

Table 7.1: Test-time harmonization at $8\times$ ($N_{\text{img}} = 24$ images, 4 posterior samples). Δ is the prior-gain of eq. (4.3); "paired vs native" is the per-image difference against the un-harmonized arm with 95% CI and exact Wilcoxon p. `ctrl` is the control: identical pipeline but histogram-matched to a *non*-BraTS reference (the shifted set's own pooled histogram).

Arm	prior	zero-fill	Δ (dB), 95% CI		paired vs `native`
`native` (none)	19.37	18.11	$+1.26$	$[-0.84, +3.35]$	—
`mask`	22.29	17.59	$+4.70$	$[+2.12, +7.28]$	$+3.44\,[+2.50, +4.39]$, $p{=}1.2\times10^{-7}$
`mask+pct`	20.42	15.43	$+4.99$	$[+2.68, +7.30]$	$+3.73\,[+2.92, +4.55]$, $p{=}1.2\times10^{-7}$
`ctrl` (non-BraTS ref)	20.70	15.74	$+4.96$	$[+3.74, +6.18]$	$+3.70\,[+2.54, +4.86]$, $p{=}1.7\times10^{-6}$

Harmonization **nearly quadruples the prior-gain**, from $+1.26$ to $+4.70$ dB (`mask`; $+4.99$ with percentile normalization, read with the caution below), and the paired improvement holds on *every one* of the 24 images ($p \approx 10^{-7}$). The pretrained prior is therefore not intrinsically useless off-domain: it was being fed inputs whose geometry and scale it had never seen.

Where the gain actually comes from. We print both absolute terms of Δ so the reader can decompose it. For `mask` the improvement is carried by the *reconstruction*: it rises $19.37 \to 22.29$ dB while the zero-filled baseline barely moves ($18.11 \to 17.59$), the cleanest evidence that the prior genuinely works better on a skull-stripped input. For `mask+pct` part of the larger Δ is earned the other way: contrast stretching penalises the blurry zero-filled image more than the sharp reconstruction (baseline falls to 15.43). We therefore treat `mask` as the headline and read the extra 0.3 dB of `mask+pct` with caution.

Which step does the work, and a control that fires. The skull-strip alone delivers almost the entire effect ($+4.70$ of $+4.99$ dB); percentile normalization adds a little; histogram matching *alone*, without a mask, is catastrophic ($\Delta = -11.80$ dB, 0% of images, $N_{\text{img}} = 16$): matching a skull-bearing image against a brain-only reference destroys the anatomy. The **control arm fires**: matching to a *non*-BraTS reference is statistically indistinguishable from no histogram matching at all (difference -0.03 dB, 95% CI $[-1.51, +1.45]$, exact Wilcoxon $p = 1.0$). We therefore *reject* the interpretation we initially expected: the benefit is **not** BraTS-specific intensity alignment. What matters is *geometry* (no skull) and *scale*; the shape of the intensity histogram is irrelevant. This was only visible because the control was pre-specified.

What this does not establish. Beyond being an oracle, the improvement is not even attributable to the *prior*: removing the skull hands the reconstructor an exactly known compact support, which any support-exploiting algorithm can convert into dB. A prior-free support-constrained baseline in fact gains *more* from the same harmonization than the score prior does. And the remedy has a hard boundary across acceleration ($N_{\text{img}} = 16$, `full` arm): the gain goes $+3.86 \to +6.84$ dB at $4\times$ and $+1.22 \to +3.81$ at $8\times$, but at $24\times$ it stays *negative* in both arms ($-2.90 \to -2.15$ dB): under extreme undersampling on shifted data the prior still does more harm than good. The defensible reading is narrow but real: *input geometry and scale cause most of the failure of section 6.2 at moderate acceleration; at $24\times$ they explain it only partially, since even the oracle input fix leaves the prior net-harmful.* Everything that follows tests deployable consequences of exactly that diagnosis.

7.2 Repair attempt 1: constraining the input (a null, and a lesson)

The deployable version of the harmonization idea estimates the brain support from the *zero-filled* image (available the moment the scan is acquired, with no ground truth touched) and enforces it *inside* the sampler, scoring against the untouched native scan. How the constraint is applied matters more than whether:

$$\begin{aligned} \text{hard:} \quad & \mathbf{x}_t \leftarrow \mathbf{m} \odot \mathbf{x}_t, \\ \text{noise-consistent:} \quad & \mathbf{x}_t \leftarrow \mathbf{m} \odot \mathbf{x}_t + (1 - \mathbf{m}) \odot \sigma(t)\, \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \end{aligned} \tag{7.1}$$

The hard variant is the naive choice and it is *wrong* for a VE-SDE: a valid iterate at time t is $\mathbf{x}_0 + \sigma(t)\mathbf{z}$ over the whole frame with σ up to 128, so zeroing the exterior removes noise the score network expects and presents a mask-shaped discontinuity far outside its training distribution. This is precisely the mistake that eq. (3.1) exists to avoid, made by us in a new place. The noise-consistent variant keeps the exterior at exactly the right noise level. A *prior-free* control (POCS [36]: alternating projection onto the measured k-space and the support, no learned model) runs alongside. Table 7.2 reports both variants plus an *oracle-support* arm whose mask comes from the clean image.

Table 7.2: Support-constrained reconstruction ($N_{\text{img}} = 24$ images, 4 posterior samples, in-brain PSNR in dB against the native scan). "paired vs none" is the per-image difference against the unconstrained sampler (t-based 95% CI, exact Wilcoxon p). The estimated mask is generous (0.62–0.84 of the frame); the oracle mask is tight (0.46).

Arm	$4\times$	$8\times$	$24\times$	paired vs none ($4\times$)
zero-filled	20.37	18.11	15.62	—
POCS, no prior	21.42	19.24	16.02	—
POCS + support	21.44	19.19	16.02	$+0.02$ (null)
unconstrained sampler	24.87	19.37	11.95	—
estimated support, hard	22.78	18.71	12.46	$-2.10\,[-2.88, -1.31]$, $p{=}10^{-5}$
estimated support, noise-cons.	24.16	19.45	12.13	$-0.71\,[-1.26, -0.16]$, $p{=}0.021$
oracle support, hard	13.33	13.72	13.10	-11.54
oracle support, noise-cons.	13.17	13.11	11.51	-11.71

Two mechanisms, separated by controls. Our first attempt used the hard variant, lost 2.10 dB at $4\times$, and we initially attributed the loss to the support contradicting the measured k-space (the skull *is* in the data). Three controls prove that explanation wrong for this arm, and we record the correction because it reversed a conclusion:

- The **prior-free POCS control** applies the same constraint with the same mask and no network, and is a null ($+0.02/-0.04/+0.00$ dB). A constraint that costs nothing without the network cannot itself cost 2.10 dB with it.
- **Fixing eq. (7.1)** recovers most of the loss: noise-consistent beats hard by $+1.39\,[+0.88, +1.89]$ dB at $4\times$ ($p = 5 \times 10^{-6}$) and $+0.74$ dB at $8\times$ ($p = 0.003$). The damage was our implementation error (masking a noisy iterate), not a property of support constraints.
- What remains is a small residual cost: after the fix the constraint still costs 0.71 dB at $4\times$ ($p = 0.021$) and is indistinguishable from no constraint at $8\times$ ($+0.08$, $p = 0.68$) and $24\times$ ($+0.18$, $p = 0.10$).

The *oracle* arm is the genuine instance of the conflict we posited. It survives the fix: a mask covering less than half the frame (0.46) deletes measured signal from everything outside it and stays catastrophic in both variants ($13.2/13.1/11.5$ dB), essentially flat across a $6\times$ change in measured data, the signature of a data-independent error floor. The mechanism is real but applies to tight masks, not the generous estimated one.

Verdict. Correctly implemented, a support constraint estimated from the data at hand neither helps nor materially hurts: a residual 0.71 dB cost at $4\times$ and no detectable effect at $8\times$/$24\times$. The mask one can estimate from a blurry image is too generous to recreate BraTS-like geometry, and a mask tight enough to do so contradicts the measurements. Improving the *input* is a dead end. The next section changes the *model*.

7.3 Repair attempt 2: adapting the prior (what works)

We continue denoising-score-matching training (eq. (2.8)) of the pretrained network for 1500 steps (batch 8, learning rate 2×10^{-4} under the recipe of section 2.5, EMA weights evaluated) on 400 images held out from the same target collection and strictly disjoint from the evaluation images. The split replays the evaluation selection and draws only from the remainder, with disjointness verified by assertion and by an exact-duplicate check. The total cost is *three minutes* of A100 time, once per site rather than per patient.

Table 7.3: Within-collection fine-tuning (1500 steps on 400 disjoint images), unconstrained sampler, $N_{\text{img}} = 24$ evaluation images. In-brain PSNR with prior-gain in brackets; "paired" is the per-image improvement over the un-adapted prior (t-based 95% CI, exact Wilcoxon: $p = 1.2 \times 10^{-7}$ at $4\times$ and $8\times$, 2.4×10^{-7} at $24\times$); "cluster" repeats the test on the 4 class means; last column: margin over prior-free POCS.

R	BraTS prior	adapted prior	paired improvement	cluster p	vs. POCS
$4\times$	24.87 $(+4.51)$	29.08 $(+8.72)$	$+4.21\,[+2.85, +5.57]$, 100%	0.016	$+7.66$
$8\times$	19.37 $(+1.26)$	25.47 $(+7.36)$	$+6.10\,[+4.60, +7.61]$, 100%	0.014	$+6.23$
$24\times$	11.95 (-3.67)	17.98 $(+2.36)$	$+6.03\,[+4.68, +7.39]$, 96%	0.006	$+1.96$

Three minutes of adaptation recovers 4–6 dB on essentially every image. The most consequential row is the last: at $24\times$ the un-adapted prior was *actively harmful* (-3.67 dB against zero filling) and after adaptation becomes helpful ($+2.36$). The adapted prior also beats prior-free POCS at every acceleration, so the gain is attributable to the *learned prior* rather than to the constraints around it, which was the objection that defeated section 7.1. Calibration improves too, most where it was worst (Spearman $0.507 \to 0.606$ at $24\times$). This single experiment has known limits: one seed, one set size, a single collection, and one near-duplicate pair in the splits (excluding it changes the paired effects by ≤ 0.08 dB). The next section eliminates each of these.

7.4 How little data, which seed, and does it cross sources?

We re-ran the headline adaptation study at $N_{\text{img}} = 48$ evaluation images and 500 reverse steps (half the sampling cost at a measured 0.46 dB loss, section 5.4; the paired contrasts below are unaffected, since both arms share the step count), then swept the adaptation set size and the training seed, tracked a held-out DSM loss throughout, and added a cross-source arm.

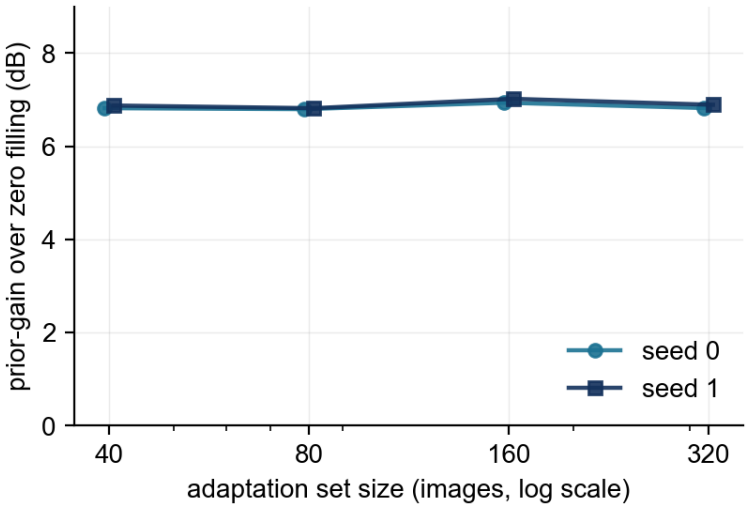


Figure 7.1: Prior-gain at $8\times$ against adaptation set size, two seeds per size (the sweep of this section, evaluated on the $N_{\text{img}} = 24$ set of section 7.3). The curve is flat: 40 images buy what 320 buy.

Table 7.4: Adaptation re-measured on $N_{\text{img}} = 48$ images at 500 reverse steps. Prior-gain in brackets; "paired" as before.

R	BraTS prior	adapted prior	paired improvement
$4\times$	23.21 ($+2.92$)	28.29 ($+8.00$)	$+5.09\,[+4.04, +6.13]$, 100%, $p{=}7\times10^{-15}$
$8\times$	17.48 (-0.59)	24.60 ($+6.53$)	$+7.12\,[+5.96, +8.29]$, 100%, $p{=}7\times10^{-15}$
$24\times$	11.40 (-4.25)	17.94 ($+2.30$)	$+6.55\,[+5.89, +7.20]$, 100%, $p{=}7\times10^{-15}$

On the doubled sample the un-adapted prior's point estimate turns negative at $8\times$ as well (-0.59 dB, harmful on exactly 24 of 48 images, 95% CI $[-2.05, +0.86]$: indistinguishable from zero), and at $24\times$ it is unambiguously harmful (-4.25 dB, 94% of images below zero filling). Adaptation flips both to clearly positive and improves *every one* of the 48 images at all three accelerations.

Forty images are enough; the seed does not matter. Figure 7.1 plots the sweep. Sweeping the adaptation set over $\{40, 80, 160, 320\}$ images with two seeds each (at $8\times$, evaluated on the $N_{\text{img}} = 24$ image set of section 7.3, so these gains are not directly comparable to the 48-image table above): seed 0 gives $+6.82/+6.80/+6.93/+6.82$ dB and seed 1 gives $+6.87/+6.82/+7.02/+6.89$ dB: mean $+6.87$, range 0.22 dB, standard deviation 0.076 across all eight runs. The curve is flat: 40 images buy the same gain as 320, and training randomness contributes an order of magnitude less variation than the effect size. In practice a site needs on the order of *forty* scans, not hundreds.

No memorisation signature. Every run tracks a DSM loss on 80 held-out images never trained on. It falls alongside the training loss at every set size, and the train/held-out gap shows no widening beyond batch noise as the training set shrinks: $+348$, $+224$, $+264$, $+200$ nats at $n = 40, 80, 160, 320$ (mean of both seeds over the last three checkpoints; the per-batch DSM loss swings by ±1000 nats, so single-checkpoint gaps are uninterpretable). Memorisation would show as the smallest set overfitting hardest; here the four gaps span

only 148 nats, small against those swings, and even the nominally largest, at $n = 40$, is well within a single batch fluctuation.

It transfers across sources. The decisive test uses the collection's internal structure: the no-tumour class comes from Br35H while the tumour classes come from figshare/SARTAJ. We adapt on 244 *tumour-class* images and evaluate on 32 *no-tumour* images: a different source *and* different content, never seen during adaptation. At $8\times$ the un-adapted prior is harmful on this unseen source (15.66 dB, gain -2.28, helping only 31% of images); after adaptation it is clearly useful (24.00 dB, gain $+6.05$, 97%). The paired improvement is $+8.33$ dB $[+6.70, +9.96]$, $p = 9 \times 10^{-10}$. What the prior learns from one source's brain MRI carries to another's within the same collection: the repair is not a within-source artefact.

7.5 Replication on a second source: OASIS-1

Every off-domain number so far comes from one aggregated collection whose images are JPEG-compressed and carry no patient identifiers, two potential confounds. OASIS-1 removes both. Its volumes are distributed in native medical format (no lossy compression), acquired on a documented scanner (1.5 T Siemens MP-RAGE), and carry `OAS1` subject identifiers, so the split can be made *subject-wise*: one central axial slice per subject, 48 evaluation, 40 adaptation and 20 held-out subjects, all disjoint; every count in this section is a count of *people*. Like the Kaggle collection, and unlike BraTS, the raw scans keep the skull and native intensities: the same pipeline shift, from a clean source.

Table 7.5: OASIS-1 replication ($N_{\text{img}} = 48$ subjects, one slice each, $S{=}4$ samples, 500 steps, in-brain PSNR). "Gain" is the prior-gain over zero filling; "Δ" the paired adapted$-$base improvement with 95% CI; all three $p = 7 \times 10^{-15}$ (every subject improves).

R	Base gain	wins	Adapted gain	wins	Δ
$4\times$	$+4.22$	100%	$+6.87$	100%	$+2.65\,[+2.51, +2.78]$
$8\times$	$+2.80$	94%	$+5.74$	100%	$+2.94\,[+2.71, +3.18]$
$24\times$	-4.22	0%	$+2.16$	98%	$+6.38\,[+6.08, +6.68]$

Figure 7.2 shows every subject. Both halves of the thesis's central claim replicate. The *failure* replicates: at $24\times$ the un-adapted prior is actively harmful on every one of the 48 subjects (-4.22 dB against zero filling, 0% wins), on data that cannot be blamed on JPEG artefacts or repeated patients. The *repair* replicates: the same 1500-step recipe (158 s on one GPU, 40 subjects) lifts every acceleration, flips $24\times$ from harmful to useful ($-4.22 \to +2.16$, 98% of subjects), and improves every subject at every acceleration ($p = 7 \times 10^{-15}$, the exact-test floor at $n = 48$). Calibration improves alongside (Spearman $0.74 \to 0.79$ at $8\times$). The failure is milder here than on the Kaggle collection at $4\times$–$8\times$ (the prior still helps before adaptation), which is itself informative: OASIS-1's native intensities sit closer to BraTS than JPEG-compressed web images do, so the size of the pipeline shift tracks the size of the harm. Two limitations stand: transfer of a single adaptation across independently curated collections (for example, a Kaggle-adapted prior evaluated on OASIS-1) was not tested, and the framing-re-fit mechanism of section 7.7 gives no reason to expect it. OASIS-1 is also a single-site collection, so multi-site replication with named scanners remains future work.

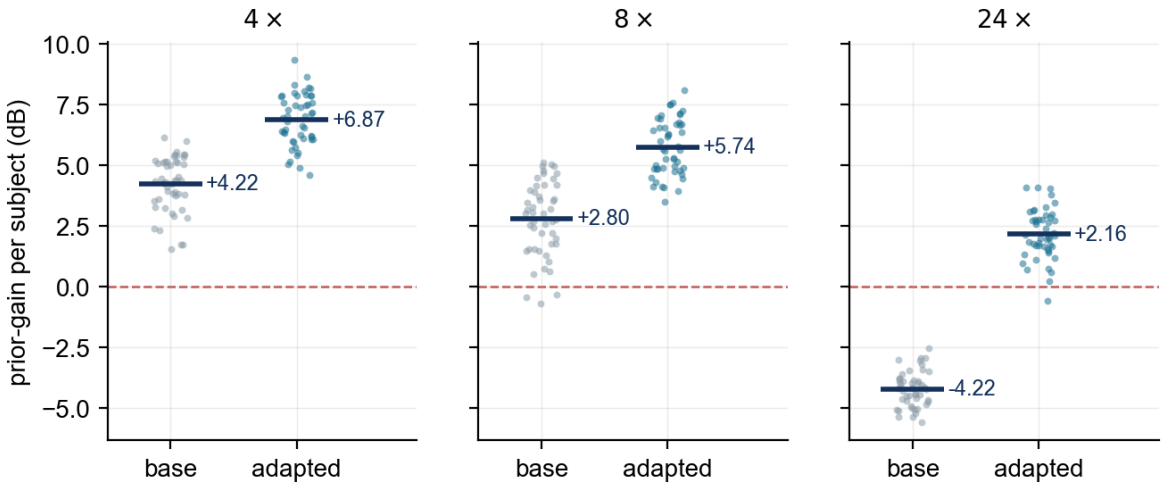


Figure 7.2: OASIS-1 replication: per-subject prior-gain before and after adaptation ($N_{\text{img}} = 48$ subjects, one slice each; the data of table 7.5). Horizontal bars mark means; the dashed line is the zero-gain level, below which the prior is harmful.

7.6 Selective prediction: making the uncertainty actionable

Finally we convert the calibrated uncertainty of section 5.2 from a property into a decision rule, evaluated with the risk–coverage protocol of section 4.2: pixels are retained in order of increasing predicted uncertainty, with random and oracle deferral bounding the achievable range.

Table 7.6: Selective prediction at $8\times$ ($N_{\text{img}} = 24$ images). Risk is in-brain RMSE over retained pixels; "vs random" is the error reduction relative to deferring the same fraction at random; AURC is the area under the risk–coverage curve (lower is better).

Prior	risk @ 100%	@ 90%	@ 50%	vs random @50%	AURC (unc./rand./oracle)
BraTS	0.1264	0.1214	0.1116	−11.7%	0.1176 / 0.1263 / 0.0723
adapted	0.0578	0.0531	0.0423	−27.0%	0.0490 / 0.0579 / 0.0302

Deferring the least reliable half of the pixels cuts the error on the remainder by 27% relative to random deferral, and by 8.3% already at 90% coverage. The uncertainty also becomes *more* useful after adaptation (−27.0% vs −11.7%): a prior that fits the data produces a disagreement signal that tracks the true error more faithfully. The oracle column bounds what better uncertainty estimation could still buy. We stress the claim's boundary: this demonstrates a property of the map (its ranking is exploitable), not a validated human-review workflow.

7.7 Why does three minutes of fine-tuning work so well?

The size of the repair (5–8 dB from 1500 gradient steps on as few as 40 images) deserves a mechanistic comment, which we offer as interpretation rather than as a measured claim. Three ingredients plausibly compound. First, the network is small (2.7M parameters), so 1500 steps at batch 8 (12,000 image presentations) constitute substantial optimisation pressure. Second, the DSM objective (eq. (2.8)) supervises *every pixel at every noise level*: each batch carries orders of magnitude more constraint than a per-image label would, which is why generative fine-tuning can be so data-efficient here. Third, and consistent

with the diagnosis of section 7.1, the adaptation does not need to teach the prior new anatomy so much as new *framing*: skulls, backgrounds, intensity scales and orientations. Re-fitting image statistics of this kind is a far easier learning problem than learning brains from scratch; hence 40 images suffice and the gain saturates immediately (fig. 7.1).

7.8 The causal chain, assembled

The chapter's experiments compose into one argument. The prior fails off-source (section 6.2); input geometry and scale cause most of the failure, though only part of it at $24\times$, and the prior's capacity is not the cause (oracle harmonization with firing control, section 7.1); it cannot be fixed from the input side at test time, because an estimable support is too loose and a tight support contradicts the measurements (section 7.2); it *is* fixed on the model side by minutes of DSM fine-tuning, with data requirements of tens of images, seed-independence, no memorisation signature, and cross-source transfer within the adaptation collection (sections 7.3 and 7.4); and the repaired prior's uncertainty is restored, and even strengthened, as a decision signal (section 7.6). Each link was established against a control that could have broken it; one control did break our own first explanation, which we report as part of the record (section 7.2).

8. RELATED WORK

8.1 Compressed sensing and classical priors

Accelerated MRI predates deep learning by a decade. Compressed sensing established that a signal sparse in some transform domain can be recovered exactly from incoherent sub-Nyquist measurements [3, 4], and Lustig et al. [5] brought the theory to MRI, where the Fourier sampling and wavelet-sparsity structure fit naturally. These methods use a *hand-crafted* prior (sparsity); their descendants replace it with a learned one, but retain the same two-term structure of data consistency plus prior. That structure is clearly visible in the method studied here, whose data-consistency step is a direct relative of the classical projection operators (POCS [36]), used in section 7.2 as the prior-free control.

8.2 Score-based and diffusion generative models

The generative machinery originates in score matching [8] and its denoising form [9], turned into a generative model by noise-conditional score networks with annealed Langevin sampling [6] and independently by denoising diffusion probabilistic models [37]. The SDE formulation of Song et al. [2] unifies both families, supplies the reverse-time SDE [11], the probability-flow ODE and the Predictor–Corrector sampler, and introduces the NCSN++ architecture, the exact backbone reused here. Practical guidance on setting $\sigma_{\max}$ from the data's pairwise distances comes from [34]; it is the reason intensity *scale* matters so much in section 7.1.

8.3 Generative priors for MRI reconstruction

Jalal et al. [7] solve compressed-sensing MRI by annealed Langevin posterior sampling under a score-based prior trained on fastMRI [38], and argue theoretically and empirically that *posterior sampling* confers robustness to distribution shift; in particular, a brain-trained prior transfers to other anatomy under their protocol. Our domain-shift results (section 6.2) complement rather than contradict theirs, and the difference is instructive. Their shifted inputs pass through the *same* acquisition and preprocessing pipeline as their training data; what changes is the anatomy. Ours change the *pipeline itself* (skull present, scale uncalibrated, orientation mixed, JPEG artefacts) while keeping the anatomy. The two axes of shift behave very differently: posterior sampling tolerates the first gracefully, but the second moves the inputs off the manifold where the score was ever trained. There we find the method brittle (to the point of harm) but repairable. Song et al. [1], the paper reproduced here, add the efficient data-consistency projection and extend the approach to CT; Chung and Ye [39] develop score-based reconstruction for accelerated MRI, training on magnitude images alone while reconstructing complex-valued and multi-coil data, agnostic to the sub-sampling pattern. The common architectural property, one unconditional prior for many measurement operators, is exactly what our uncertainty and adaptation studies exploit.

8.4 Uncertainty quantification in reconstruction

Posterior sampling provides uncertainty "for free" as sample dispersion [7]; Edupuganti et al. [40] study uncertainty in deep MRI reconstruction using variational autoencoders

and Stein estimators. Our contribution is not a new estimator but a *measurement*: the calibration of the posterior-sample standard deviation against the realised error for this specific solver (Spearman 0.81–0.87 in distribution, degrading to 0.31–0.67 off-source), plus its conversion into a selective-prediction rule [26] with quantified gains over random deferral (section 7.6).

8.5 Robustness, instabilities and hallucination

Antun et al. [30] demonstrated that deep reconstruction networks can be unstable: small perturbations or slight distribution changes produce large artefacts; Bhadra et al. [31] formalised *hallucination* in tomographic reconstruction as the fabrication of plausible but false structure; radiologist evaluation in the 2020 fastMRI challenge documented hallucinations in finalist submissions and named their detection a gap in its quantitative analysis [32]. Darestani et al. [41] measured robustness of compressive-sensing methods under distribution shift, finding degradation across both classical and learned methods. This literature motivates our failure-mode chapter directly. Our synthetic-lesion sweep (section 6.5) brackets an *erasure regime* (feature size $\times$ acceleration) for this particular solver, and our real-tumour comparison (section 6.3) tests the erasure worry on genuine pathology.

8.6 Domain adaptation for reconstruction priors

Our repair result, in which forty images and three GPU-minutes of DSM fine-tuning restore a mismatched prior (section 7.4), is an instance of test-domain adaptation, related in spirit to intensity standardisation pipelines (Nyúl harmonization [35], skull-stripping tools such as SynthStrip [33]) on the input side, and to transfer learning on the model side. The controlled comparison of the two routes (section 7.2 vs section 7.3) is, to our knowledge, the empirically interesting part: for this solver, adapting the *model* is cheap and decisive, while constraining the *input* at test time is neutral at best.

8.7 A contrasting paradigm: discriminative foundation models

Industrially, a prominent direction in medical-imaging AI is not generative reconstruction but *discriminative* foundation models: Curia [42] is a CT+MRI vision foundation model trained self-supervised on large exam volumes for classification, detection, staging and prompted segmentation, with the promptable tumour model ONCOPILOT [43] as a companion. The two paradigms are complementary: a foundation model learns *representations* of already-reconstructed images, whereas the score-based prior *produces* the image from partial measurements. A natural bridge is left for future work (chapter 9): reconstruct with a generative prior, then hand the uncertainty-annotated image to a discriminative model that knows where not to trust it.

9. CONCLUSIONS

9.1 Summary of findings

This thesis reimplemented the score-based inverse-problem solver of Song et al. [1] from scratch in PyTorch, reusing the authors' pretrained NCSN++ prior through a verified weight conversion. It then subjected the method to the question its publication leaves open: how far it can be trusted, and what it costs to restore that trust.

The findings are quantified in full where they are established (sections 5.1, 5.2, 6.2, 7.1 and 7.4); here we give only a qualitative summary. **The method reproduces**: agreement with the published table within 0.15 dB, across a framework port, a 473-tensor weight conversion and a re-derived measurement pipeline (section 5.1). **Its stochasticity is an asset**: averaging repeated runs beats every published single-run number, the disagreement between runs ranks pixels by their true error, and that ranking supports selective review. Half the sampling steps buy essentially the same quality (sections 5.2, 5.4 and 7.6). **Off its home distribution it can be worse than nothing**: on independently sourced brain MRI the prior's gain over zero filling collapses, turning negative at $24\times$ for 94–100% of images across both sources, and at $8\times$ on a source- and protocol-dependent subset, while its reliability signal weakens at the same time (sections 6.2 and 7.5). Real tumours are not selectively erased (section 6.3), but a small, strongly undersampled lesion can vanish, and a faint one vanishes without being flagged (section 6.5). **The failure is diagnosable and cheaply repairable**: controlled experiments attribute most of it to input geometry and intensity scale, though at $24\times$ the attribution is partial (section 7.1); constraining the input turns out neutral at best once a bug our own controls exposed is fixed (section 7.2). Three GPU-minutes of fine-tuning repair the prior on every test image; roughly forty target images suffice, and the repair carries across sources within the adaptation collection (sections 7.3 and 7.4). Both the failure and the repair replicate on a second, natively formatted source under subject-wise splits (section 7.5), and real tumours under expert masks show the reconstruction is least faithful at the pathology, with the uncertainty correspondingly elevated there (section 6.4). **Negative results are results**: guided acquisition does not beat random under a properly powered paired test, and the encouraging preliminary number was noise (section 5.3).

9.2 What this means

For the practitioner, the thesis supports three concrete recommendations. First, *never* deploy a pretrained reconstruction prior on a new site without measuring its gain over zero filling; the sign of that gain is not guaranteed. Second, if the gain is poor, the cheapest effective intervention is a minutes-long fine-tune on a few dozen site-local images, not input preprocessing. Third, run the sampler several times: the average is a better image, and the disagreement map (a ranking, not a guarantee) is informative enough to direct review, and becomes more informative the better the prior fits.

For the methodologist, the thesis shows how much inexpensive controls buy: paired designs, pre-specified controls, prior-free baselines and adequate sample sizes repeatedly changed the conclusion. They falsified our own first explanation of the support-constraint result, exposed an implementation bug, deflated a spurious adaptive-acquisition win, and prevented an oracle study from being mistaken for a deployable method. None of these

instruments cost more than minutes of compute.

For reproduction studies specifically, two lessons generalise. A faithful reimplementation is itself a severe end-to-end test: agreement within 0.15 dB across a framework port and a 473-tensor weight conversion leaves essentially no room for silent errors in the validated solver path, a stronger correctness guarantee than any unit-test suite alone; the one flagged baseline discrepancy (section 5.1) shows the same comparison working as a detector. And the most informative experiments were the ones designed to be able to *fail*: the pre-specified control arm that fired (section 7.1), the prior-free baseline that zeroed out a mechanism we believed in (section 7.2), and the held-out loss that could have revealed memorisation and did not (section 7.4).

9.3 Future work

Three directions follow directly. (i) *Clinical validation*: the expert-mask correlation of section 6.4 is a first quantitative step; what remains is testing the selective-prediction workflow with human readers, which would move the map's status from ranking to safety evidence. (ii) *Harder acquisition settings*: multi-coil complex k-space with measured noise (e.g. fastMRI [38]), prospective undersampling, and per-scan adaptive acquisition, in which a sharper importance signal than eq. (3.4) may yet beat random. (iii) *Cheaper adaptation*: given that 40 images and 1500 steps suffice, how small can the update itself be (adapters, low-rank updates, or normalisation-only fine-tuning), and can adaptation run *online* from the reconstructions themselves? A fourth, longer-range direction is the bridge of chapter 8: feeding uncertainty-annotated reconstructions to discriminative foundation models, so that downstream decisions inherit the reconstruction's own honesty about where it guessed.

TABLE OF TERMINOLOGY

English term	Ελληνικός όρος
inverse problem	αντίστροφο πρόβλημα
ill-posed	κακώς ορισμένο
prior (distribution)	εκ των προτέρων κατανομή
posterior (distribution)	εκ των υστέρων κατανομή
score (function)	συνάρτηση σκορ
score-based generative model	μοντέλο παραγωγής βασισμένο σε σκορ
diffusion model	μοντέλο διάχυσης
stochastic differential equation	στοχαστική διαφορική εξίσωση
denoising score matching	αντιστοίχιση σκορ μέσω αποθορυβοποίησης
undersampling	υποδειγματοληψία
acceleration factor	παράγοντας επιτάχυνσης
k-space	χώρος k (χώρος χωρικών συχνοτήτων)
data consistency	συνέπεια δεδομένων
zero-filled reconstruction	ανακατασκευή με μηδενική συμπλήρωση
uncertainty quantification	ποσοτικοποίηση αβεβαιότητας
calibration	βαθμονόμηση
distribution shift	μετατόπιση κατανομής
out-of-distribution	εκτός κατανομής (εκπαίδευσης)
fine-tuning	μικρο-προσαρμογή
skull-stripping	αφαίρεση κρανίου
histogram matching	αντιστοίχιση ιστογράμματος
hallucination	ψευδαίσθηση (επινόηση δομής)
lesion	βλάβη
selective prediction	επιλεκτική πρόβλεψη
memorisation	απομνημόνευση
paired test	ζευγαρωτός έλεγχος
null result	μηδενικό (αρνητικό) αποτέλεσμα
confidence interval	διάστημα εμπιστοσύνης
gradient descent	κατάβαση κλίσης
learning rate	ρυθμός μάθησης
gradient clipping	ψαλιδισμός κλίσης
exponential moving average	εκθετικός κινητός μέσος

ABBREVIATIONS – ACRONYMS

ALD	Annealed Langevin Dynamics
AURC	Area Under the Risk–Coverage curve
BraTS	Brain Tumor Segmentation (challenge/dataset)
CC0/CC BY	Creative Commons (Zero / Attribution) licence
CDF	Cumulative Distribution Function
CI	Confidence Interval
CT	Computed Tomography
DSM	Denoising Score Matching
EMA	Exponential Moving Average
FFT	Fast Fourier Transform
GPU	Graphics Processing Unit
JPEG	Joint Photographic Experts Group (lossy image format)
LIDC	Lung Image Database Consortium
MAD	Median Absolute Deviation
MCMC	Markov Chain Monte Carlo
MMSE	Minimum Mean Squared Error
MRI	Magnetic Resonance Imaging
MSE	Mean Squared Error
NCSN	Noise-Conditional Score Network
NKUA	National and Kapodistrian University of Athens
ODE	Ordinary Differential Equation
OOD	Out Of Distribution
PC	Predictor–Corrector (sampler)
POCS	Projection Onto Convex Sets
PSNR	Peak Signal-to-Noise Ratio
RMSE	Root Mean Squared Error
ROI	Region Of Interest
SDE	Stochastic Differential Equation
SSIM	Structural Similarity Index Measure
VE	Variance Exploding (SDE)

ANNEX I: HYPERPARAMETERS AND CONFIGURATION

Every experiment in this thesis was run with the configuration below; deviations are stated where the experiment is reported. The sampling hyperparameters are the tuned optima published by Song et al. [1]; the optimisation hyperparameters are those of the pretrained prior's original training recipe, kept identical during fine-tuning so that the comparison is fair.

Table A.1: Model, sampler and optimiser configuration.

Group	Setting	Value
Model	architecture	NCSN++ (time-conditional U-Net; $2{,}676{,}069$ parameters)
	input size	240×240, single channel, values in $[0, 1]$
	checkpoint	authors' EMA weights, `checkpoint_26` (converted Flax$\rightarrow$PyTorch)
SDE	type	Variance Exploding
	σ_{min}, σ_{max}	0.01, 128
	schedule	$\sigma(t) = \sigma_{\text{min}}(\sigma_{\text{max}}/\sigma_{\text{min}})^t$, $t \in [0, 1]$
Sampler	type	Predictor–Corrector (reverse diffusion + Langevin)
	steps N	1000 (default) or 500 (speed setting, section 5.4)
	corrector SNR η	0.577
	data-consistency λ	0.982 during sampling; 1.0 at the final denoise
Optimiser (fine-tune)	type	Adam (β_1=0.9, β_2=0.999, δ=10^{-8})
	learning rate	2×10^{-4} with linear warm-up over 5000 steps
	gradient clipping	global ℓ_2 norm ≤ 1
	EMA decay	0.999 (capped at $(1+n)/(10+n)$ early in training)
	steps / batch	1500 / 8

ANNEX II: REPRODUCIBILITY

The thesis is accompanied by a self-contained repository with: (i) the PyTorch reimplementation (`scoreinv/`: SDE, samplers, measurement operators, data-consistency solvers, uncertainty aggregation, support constraints, selective-prediction utilities, harmonization); (ii) the checkpoint-conversion script (Flax$\rightarrow$PyTorch, 473 tensors verified by name and shape); (iii) the experiment drivers used on the GPU instances; (iv) the raw result files (JSON) from every run, from which each table in this document is generated programmatically; and (v) unit tests covering the FFT round-trip, mask construction, and the data-consistency step at $\lambda{=}1$. All datasets used are public: the authors' released BraTS evaluation file, the Brain Tumor MRI Dataset (CC0), and LIDC chest CT. No clinical or private data were used. Reported statistics use paired designs wherever the comparison permits, exact Wilcoxon signed-rank tests, t-based 95% confidence intervals, and Welch tests for unpaired comparisons; the statistical protocol is defined once in section 4.3 and reused throughout.

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